Talk Less, Work More: Communication-Efficient Decentralized Stochastic Approximation

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Abstract

Decentralized Stochastic Approximation (SA) enables cooperative fixed-point solving over mesh networks without a central server. In these systems, local update is relatively cheap compared with coordination, and frequent information exchange makes communication cost the main bottleneck. To alleviate this issue, we propose a decentralized SA method in which each agent performs H local updates before each communication round. By amortizing communications over multiple local SA updates, the method can substantially reduce communication cost. We also study a multiple mixing scheme based on FastMix, which trades a small amount intra-round communications for a stronger consensus effectiveness on poorly connected graphs. Theoretically, we establish finite-time bounds under general contractive norms and Markovian sampling, by combining contraction, consensus error, and delayed Markovian terms in a round-level Lyapunov recursion. The results show trade-offs among computation, communication and sampling. Numerical experiments on decentralized Q-learning, Markovian SGD and linear SA validate the theory and show substantial communication savings for target accuracy.

1 Introduction

Stochastic Approximation (SA) is a fundamental approach for solving fixed point and root finding problems when the underlying operator is only observed through noisy samples [24, 19]. It has demonstrated wide applications, such as stochastic optimization [6], reinforcement learning [3, 29, 31], game theory [15] and dynamical system [18].

In modern deployments, data are distributed across multiple agents connected by a network without a central coordinator. This motivates decentralized stochastic approximation [4], where m agents cooperatively solve a fixed point equation over a mesh network, which is defined as:

$$x = \bar{F}(x) := \frac{1}{m}\sum_{i=1}^{m} F_i(x), \tag{1}$$

where $F_i : \mathbb{R}^d \to \mathbb{R}^d$ is the local operator only accessed by agent i. Existing approaches for solving this problem is to alternate one stochastic approximation update with one communication step [17, 38]. In practice, communication cost is expensive compared with local computation. Full-precision update exchange in large-scale distributed training could be sizable per iteration, making communication a bottleneck [1, 21, 5]. The bottlenecks of communication also arise in federated learning, where limited device bandwidth and energy budgets, together with intermittent connectivity, further hinder scalability [16]. To reduce communication overhead, local-update strategies have been widely adopted, allowing agents to perform multiple local steps between communication

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Table 1: Comparison of settings for decentralized stochastic approximation.

Work	Decentralized	Multi-local-step	Markovian	General norm	Avg-only contraction	Non-asymptotic	Fast mix
DSA [4]	✓	✗	✗	✗	✓	✗	✗
Biased DSA [32]	✓	✗	✓	✗	✓	✓	✗
DCSA [38]	✓	✗	✓	✗	✓	✓	✗
LSA [13]	✗	✓	✓	✗	✓	✓	–
DLMSA [33]	✓	✓	✓	✓	✗	✓	✗
DLSA (ours)	✓	✓	✓	✓	✓	✓	✓

rounds. Representative examples of local-update methods include local SGD [30] and its federated counterparts [22]. By performing multiple local steps between communication rounds, these methods can substantially reduce synchronization frequency compared with fully synchronous training.

Terminology. Here "average-only contraction" means that only the averaged operator $\bar{F}$ is assumed contractive; the local mean operators $\bar{F}_i$ may be noncontractive. We use "finite-time" and "non-asymptotic" synonymously.

In decentralized training with no central node for aggregation, multiple communication steps is usually required in each communication round, especially when the network connectivity is poor. In all existing works on decentralized stochastic approximation [4, 32, 38, 33], only single gossip communication step is adopted per communication round, making their decentralized algorithms not robust to poorly-connected networks. Beyond communication bottlenecks, another challenge in solving (1) is that the averaged operator $\bar{F}$ is contractive only under a problem-dependent, non-Euclidean norm. Conventional examples exist in tabular reinforcement learning, where Bellman operators are naturally contractive in the ℓ_∞ norm [10] and non-trivial norm [34]. However, existing local stochastic approximation results do not yet provide a complete theory for fully decentralized settings under non-Euclidean norm contractions as shown in Table 1.

In this work, we develop a communication-efficient decentralized stochastic approximation method with local updates that explicitly addresses these issues. Our Decentralized Local Stochastic Approximation (DLSA) algorithm targets the setting in which the averaged operator is contractive with respect to an arbitrary norm, while allowing multiple local updates. Our DLSA composes of two levels in communication. The higher level is communication round which happens after every H local updates. A large H can save communication costs by reducing the communication frequency. Within each communication round, there is an optional procedure when the network is poorly connected: rather than one-step gossip which is too weak to effectively mix the information, we adopt an accelerated multi-consensus routine based on FastMix, thereby improving the effective mixing per each communication round. This perspective allows us to study communication efficiency rather than only iteration complexity, which is the dominant focus of most existing works.

On the theory side, we use a generalized Moreau envelope to construct a potential function compatible with the contraction norm and embed it into a Lyapunov function that also tracks consensus error and a delayed term induced by Markovian sampling. The results demonstrate tradeoffs among communication, computation and sampling: i) large H reduces communication frequency and cost, but may enlarge the ultimate error ball under steady state, ii) a high-quality (resp. poor) network condition provides more (resp. less) tolerance on a worse mixing rate of Markovian sampling.

Contributions

- We propose DLSA, a decentralized stochastic approximation algorithm that performs multiple local SA updates per communication round, yielding a direct reduction in communication frequency while approaching the target fixed point.
- For poorly-connected network where one-step gossip is insufficient in each communication round, we adopt an accelerated multi-consensus routine based on FastMix to enhance the communication, making our algorithm robust to the quality of network.
- We develop a finite time analysis for DLSA when only the averaged operator is contractive under an arbitrary norm, covering non Euclidean contractions such as the infinity norm in tabular Q-learning or some nontrivial norm in averaged reward TD learning.
- We evaluate DLSA on various settings, including linear stochastic approximation, Markovian stochastic gradient descent and reinforcement learning, and demonstrate substantial communication savings.

2 Related works

Distributed stochastic approximation typically combines local stochastic updates with gossip communications [7], so that all agents approach a common fixed point of the averaged operator with converged consensus error. An asymptotic convergence analysis of decentralized SA with i.i.d. distribution of the noise in operator is derived in [4], while finite-time theory of decentralized SA under Markovian noise is more preferred for practice. Non-asymptotic analysis under Markovian noise was relatively underexplored even in the centralized setting, until recently: [3] and [29] study linear SA, while [23, 11] focus on nonlinear SA.

Building on these developments, finite-time analysis of decentralized SA with Markovian noise has been studied in [12, 14, 20, 38, 33]. Specifically, [12, 14, 20] study linear decentralized SA in the context of temporal-difference learning. The closest nonlinear references are [38] and [33]. The former studies one-local-step-one-communication decentralized SA in Euclidean geometry. The latter studies the same local-update template as ours, but under the substantially stronger assumption that every local mean operator $\bar{F}_i$ is contractive.

Most existing decentralized SA results focus on iteration complexity rather than communication complexity [8, 26]. Some works discuss communication costs, especially in federated settings [37, 27], but in decentralized computing with poorly connected network, multi-consensus mixing in each communication round is usually required to improve the network connectivity. The communication complexity additionally depends on the connectivity of networks and is hence not equivalent to iteration complexity. Polynomial-gossip schemes such as FastMix [35, 36] and Chebyshev acceleration [2, 25] can accelerate the multi-consensus routine and improve the network dependence. To the best of our knowledge, the decentralized SA works in Table 1 use plain gossip or centralized averaging [4, 32, 38, 13, 33], non-robust to poorly connected networks. Our contribution is to apply FastMix to decentralized local SA with Markovian sampling and general contractive norms, as well as provide the analysis on communication complexity.

3 Problem settings

This work studies finite-sample decentralized stochastic approximation over a network of m agents connected by a static, connected, undirected graph, without a central coordinator. The goal is to solve for the fixed point $x^\star \in \mathbb{R}^d$ of the averaged operator

$$x^\star = \bar{F}(x^\star), \qquad \bar{F}(x) := \frac{1}{m}\sum_{i=1}^{m} \bar{F}_i(x), \tag{2}$$

where $\bar{F}_i(x) := \mathbb{E}_{Y_i \sim \mu_i}\left[F_i(x, Y_i)\right]$, $F_i : \mathbb{R}^d \times \mathcal{Y} \to \mathbb{R}^d$, and $\mathcal{Y}$ denotes sample space with distribution μ_i. For each agent $i \in \{1, \cdots, m\}$, let $\{Y_i^{(s)}\}_{s\geq 0}$ denote the underlying Markov chain. During round k and inner step t, Algorithm 1 uses the sample $Y_{i,t}^k := Y_i^{(kH+t)}$. For any $\tilde{\epsilon} > 0$, denote by $\tilde{\tau}_{\tilde{\epsilon}}$ the smallest integer such that for all $y_i \in \mathcal{Y}$,

$$d_{TV}\left(P\Big(Y_i^{(s)} = \cdot \mid Y_i^{(0)} = y_i\Big), \mu_i\right) \leq \tilde{\epsilon}, \qquad \forall s \geq \tilde{\tau}_{\tilde{\epsilon}}. \tag{3}$$

Here d_{TV} denotes total variation distance. In addition, we assume $\mathcal{Y}$ is finite and each chain $\{Y_i^{(s)}\}_{s\geq 0}$ is irreducible and aperiodic. Then there exist constants $r_p \in (0, 1)$ and $c_p > 0$ such that

$$d_{TV}\left(P\Big(Y_i^{(s)} = \cdot \mid Y_i^{(0)} = y_i\Big), \mu_i\right) \leq c_p r_p^s, \qquad \forall s \geq 0, \tag{4}$$

and therefore $\tilde{\tau}_{\tilde{\epsilon}}$ is bounded above by $\tau_{\tilde{\epsilon}} := \frac{1}{1-r_p}\log\frac{c_p}{\tilde{\epsilon}}$.

Besides, we have the following assumptions on the operator and network.

Assumption 1 (Operator)**.** *Assume the following:*

(i) There exists $L > 0$, such that for any $x_1, x_2 \in \mathbb{R}^d, y \in \mathcal{Y}$, we have

$$\max\{\|F_i(x_1, y) - F_i(x_2, y)\|_c, \|F_i(x_1, y) - x_1 - F_i(x_2, y) + x_2\|_c\} \leq L\|x_1 - x_2\|_c, \tag{5}$$

(ii) There exists some $\beta \in (0,1)$ such that for any $x, y \in \mathbb{R}^d$,

$$\|\bar{F}(x) - \bar{F}(y)\|_c \leq \beta \|x - y\|_c. \tag{6}$$

(iii) There exists a universal constant $B > 0$, such that for any $i \in [m], y \in \mathcal{Y}$,

$$\|F_i(x^\star, y)\|_c + \|x^\star\|_c \leq B. \tag{7}$$

Here $\|\cdot\|_c$ denotes the norm of interest. Assumption 1 (ii) guarantees that $x^\star$ exists and is unique.

Remark 1 (Why average-only contraction is weaker)**.** *The assumption that only $\bar{F}$ is contractive is strictly weaker than assuming each local mean operator $\bar{F}_i$ is contractive. A one-dimensional example suffices. Let $m = 2$, $\|\cdot\|_c = |\cdot|$, and*

$$\bar{F}_1(x) = 2x, \qquad \bar{F}_2(x) = -\frac{3}{2}x.$$

Neither local operator is contractive, since

$$|\bar{F}_1(x) - \bar{F}_1(y)| = 2|x - y|, \qquad |\bar{F}_2(x) - \bar{F}_2(y)| = \frac{3}{2}|x - y|.$$

However, their average is contractive with factor $\beta = 1/4$, since

$$\bar{F}(x) = \frac{1}{2}(\bar{F}_1(x) + \bar{F}_2(x)) = \frac{1}{4}x,$$

Assumption 2 (Network)**.** *There is a mixing matrix $W = (w_{ij})_{1 \leq i,j \leq m} \in \mathbb{R}^{m \times m}$, such that*

(i) Compliance with graph: $w_{ij} > 0$ if agents i and j are directly connected in the network; otherwise, $w_{ij} = 0$.

(ii) Symmetry: $W^\top = W$.

(iii) Doubly stochastic: $W\mathbf{1} = \mathbf{1}$, $\mathbf{1}^\top W = \mathbf{1}^\top$, where $\mathbf{1} \in \mathbb{R}^m$ is an all-one vector.

Denote network parameter $\rho := \left\|W - \frac{1}{m}\mathbf{1}\mathbf{1}^\top\right\|_2$. The larger ρ is, the worse the network is connected.

Notations: Use $[m]$ to denote $\{1, \cdots, m\}$. Stacked variables $X \in \mathbb{R}^{md}$ are defined as $X := [x_1; \cdots; x_m]^\top \in \mathbb{R}^{md}$, where $x_i \in \mathbb{R}^d$ is local copy at agent $i \in [m]$. Denote the averaged variable $\bar{x} := \frac{1}{m}\sum_{i=1}^m x_i$. Denote I_d as the identity matrix with dimension d. Use $\otimes$ to denote the Kronecker product. Use u_{2c} and ℓ_{2c} to denote norm equivalence between $\|.\|_2$ and $\|.\|_c$: for any $x \in \mathbb{R}^d$, it holds $\ell_{2c}\|x\|_2 \leq \|x\|_c \leq u_{2c}\|x\|_2$ and $\delta := \frac{u_{2c}}{\ell_{2c}}$.

4 Algorithm

We present our Decentralized Local Stochastic Approximation (DLSA) method in this section. DLSA alternates between local computations and network communications. Specifically, after each communication round k, every agent starts from its current outer iterate x_i^k and performs $H \geq 1$ local stochastic approximation updates using its own Markov sample stream, which can be viewed as a stochastic inexact fixed-point iteration where the local mean operator $\bar{F}_i$ is estimated through samples of F_i. After these H local steps, agents communicate their iterates with neighbors via the mixing matrix W to form the next outer iterate x_i^{k+1}. The details of the algorithm are listed in Algorithm 1.

The communication operator $\mathcal{C}(\cdot)$ can be instantiated in two ways. For plain DLSA, we set

$$\mathcal{C}(Z_H^k) = (W \otimes I_d) Z_H^k,$$

Algorithm 1 Decentralized Local Stochastic Approximation (DLSA)

Input: $\mathbf{x}^0 = (x_i^0)_{i \in [m]}$
Output: $\mathbf{x}^{K+1}$
Parameters: stepsize $\alpha > 0$; local-update length $H \in \mathbb{N}$; communication operator $\mathcal{C}(\cdot)$
for $k = 0, \ldots, K$ **do**
  // Local stochastic approximation steps
  $z_{i,0}^k = x_i^k, \quad \forall i \in [m]$
  for $t = 0, \ldots, H-1$ **do**
    $z_{i,t+1}^k = z_{i,t}^k + \alpha\big(F_i(z_{i,t}^k, Y_{i,t}^k) - z_{i,t}^k\big), \quad \forall i \in [m]$
  end for
  // Communication step
  $X^{k+1} = \mathcal{C}(Z_H^k)$
end for

which corresponds to one gossip step. For the accelerated variant DLSA-FastMix, we set

$$\mathcal{C}(Z_H^k) = \textsc{FastMix}(Z_H^k, W, n_\rho, \eta_w),$$

where FASTMIX is given in Algorithm 2.

When the network is poorly connected ($\rho \approx 1$), multiplying mixing matrix W only one time cannot efficiently drive all agents towards consensus, and multiple times of mixings in each communication round is necessary. This can be done either by repeated mixing or, more efficiently, by an accelerated average-preserving polynomial filter implemented through FastMix.

To show the effect of FastMix, consider that simply repeating mixing n_ρ times would reduce the network parameter from ρ to ρ^{n_ρ}, in other words, to achieve some threshold of connectivity when ρ is close to one, the plain repeated mixing typically needs $n_\rho = \mathcal{O}\big((1-\rho)^{-1}\big)$ mixing steps. In contrast, FastMix improves this dependence to $\mathcal{O}\big((1-\rho)^{-1/2}\big)$.

Algorithm 2 $\textsc{FastMix}(Z^0, W, n_\rho, \eta_w)$

Input: stacked iterate $Z^0 \in \mathbb{R}^{md}$; mixing matrix W; number of gossip steps n_ρ; momentum parameter η_w;
Output: $Z^{n_\rho+1}$;
Initialize: $Z^1 = Z^0$;
for $\nu = 1, \dots, n_\rho$ **do**
  $Z^{\nu+1} = (1+\eta_w)(W \otimes I_d)Z^\nu - \eta_w Z^{\nu-1}$
end for
Return $Z^{n_\rho+1}$

More specifically, FastMix improves the quality of each communication phase by producing a polynomial filter $\widetilde{W} = P_{n_\rho}(W) \in \mathbb{R}^{m\times m}$ that approaches $\frac{1}{m}\mathbf{1}\mathbf{1}^\top$ much faster on poorly connected networks. Given an input $Z^0 \in \mathbb{R}^{m\times d}$, with $\eta_w = \frac{1}{1+\sqrt{1-\rho^2}}$, the output $Z^{n_\rho} = P_{n_\rho}(W)Z^0$ satisfies

$$\|Z^{n_\rho} - \bar{z}^{n_\rho} \otimes \mathbf{1}\| \le \sqrt{14}\Big(1 - (1-1/\sqrt{2})\sqrt{1-\rho}\Big)^{n_\rho} \|Z^0 - \bar{z}^0 \otimes \mathbf{1}\|, \tag{8}$$

so we define the effective FastMix factor

$$\rho_{\rm fm} := \left\|\widetilde{W} - \frac{1}{m}\mathbf{1}\mathbf{1}^\top\right\|_2 = \sqrt{14}\Big(1 - (1-1/\sqrt{2})\sqrt{1-\rho}\Big)^{n_\rho}.$$

Remark 2. *Given a target threshold $\bar{\rho} \in (0,1)$, it suffices to choose*

$$n_\rho \ge \frac{1}{(1-1/\sqrt{2})\sqrt{1-\rho}} \log\frac{\sqrt{14}}{\bar{\rho}} = \mathcal{O}\Big((1-\rho)^{-1/2}\Big).$$

Then $\rho_{\rm fm} \le \bar{\rho}$. Therefore, DLSA-FastMix can secure a good quality of communication with an additional communication cost that scales as $1/\sqrt{1-\rho}$ up to logarithmic factors.

4.1 Main Results

For convenience, define the consensus error $S_k := \frac{1}{m}\sum_{i=1}^m \|x_i^k - \bar{x}^k\|_c$ and $S_{k,t} := \frac{1}{m}\sum_{i=1}^m \|z_{i,t}^k - \bar{z}_t^k\|_c$, and the estimation gap $R_k := \big\|\bar{x}^k - x^\star\big\|_c$ and $R_{k,t} := \|\bar{z}_t^k - x^\star\|_c$. We now show the convergence results of Algorithm 1 as in Theorem 1, where for clearer presentation we hide and leave more details about parameters into Appendix A.4.

Theorem 1 (Finite-time convergence of DLSA)**.** *Suppose Assumptions 1 and 2 hold, and suppose the Markovian sampling satisfies the mixing condition* (4)*. Run Algorithm 1 with constant stepsize α and local-update length H. Let $\rho_{\rm eff}$ denote the effective network mixing factor in one communication round. For plain DLSA, $\rho_{\rm eff} = \rho$; for DLSA-FastMix, $\rho_{\rm eff} = \rho_{\rm fm}$. Suppose*

$$\alpha \le \frac{c_\alpha}{H(\tau+1)(1+L)^2} = \widetilde{O}\left(\frac{1}{H\tau(1+L)^2}\right), \tag{9}$$

$$\rho_{\rm eff} \le \frac{c_\rho}{\sqrt{m}\,(\tau+1)} = \widetilde{O}\left(\frac{1}{\sqrt{m}\tau}\right), \tag{10}$$

where $c_\alpha, c_\rho > 0$ are problem-dependent constants, $\widetilde{\mathcal{O}}(\cdot)$ hides logarithmic factors and some problem-dependent constants, τ denotes Markov delay and satisfies $\tau = \widetilde{\mathcal{O}}\Big(\frac{1}{H(1-r_p)}\Big)$, then for any $k \ge \tau$,

$$\begin{aligned}\mathbb{E}\big[R_k^2\big] \le{}& C_R(1-c_0\alpha H)^{k-\tau}\big(\mathbb{E}\big[R_\tau^2\big] + \mathbb{E}\big[S_\tau^2\big] + \mathbb{E}\big[S_0^2\big]\big) \\ &+ \widetilde{\mathcal{O}}\big(\alpha B^2\big(1 + H + \alpha H^2(1+L)^2 + \alpha^2 H^3(1+L)^2\big)\big),\end{aligned} \tag{11}$$

$$\mathbb{E}\big[S_k^2\big] \le C_S\,(1-c_0\alpha H)^{k-\tau}\left(\mathbb{E}\big[R_\tau^2\big]+\mathbb{E}\big[S_\tau^2\big]+\mathbb{E}\big[S_0^2\big]\right) + \widetilde{\mathcal{O}}\left(\rho_{\text{eff}}^2\alpha^2H^2B^2\left(1+\alpha(1+L)^2\left(1+H+\alpha H^2(1+L)^2+\alpha^2H^3(1+L)^2\right)\right)\right), \tag{12}$$

where $c_0, C_R, C_S > 0$ are constants independent of α, H, B and ρ_{eff}.

Theorem 1 shows that both the estimation error R_k^2 and the consensus error S_k^2 contract geometrically up to steady-state error terms. The result highlights two key tradeoffs.

(i) The theorem makes the tradeoff in H explicit: larger H improves the transient contraction factor through $\alpha\kappa H$, but the error ball typically grows polynomially with H. Thus larger H can enlarge the steady-state error ball unless the stepsize is reduced accordingly.

(ii) Algorithm 1 actually involves two mixing processes: one is consensus mixing through the physical network with quality measured by ρ_{eff}; another one is the mixing of the Markovian sampling process, whose quality is reflected by τ (small τ is wanted). The condition (10) implicitly implies a trade-off relationship between these two mixings: the better quality network tolerates a worse sampling process, and vice versa.

To achieve ε-precision for estimation error and consensus error, we have following results on communication and sample complexities, by simply applying Theorem 1.

Corollary 1 (Communication and sample complexity)**.** *Fix some small enough $\varepsilon \in (0,1)$. With the same setting as in Theorem 1, denote $\alpha_{\max} := \frac{c_\alpha}{H(\tau+1)(1+L)^2}$, and choose a constant stepsize $\alpha = \min\left\{\alpha_{\max}, \frac{\varepsilon}{B^2(1+H)}, \frac{\sqrt{\varepsilon}}{BH}\right\}$. Then, with total iterations $K = \widetilde{\mathcal{O}}\left(\tau + \frac{1}{\alpha H}\log\frac{1}{\varepsilon}\right)$, we obtain*

$$\mathbb{E}\big[R_K^2\big] \le \varepsilon, \quad \mathbb{E}\big[S_K^2\big] \le \varepsilon$$

The communication and sample complexities are respectively

$$\widetilde{\mathcal{O}}\left(\tau + \max\left\{(\tau+1)(1+L)^2, \frac{B^2(1+H)}{H\varepsilon}, \frac{B}{\sqrt{\varepsilon}}\right\}\log\frac{1}{\varepsilon}\right), \tag{13}$$

and

$$\widetilde{\mathcal{O}}\left(\tau H + \max\left\{H(\tau+1)(1+L)^2, \frac{B^2(1+H)}{\varepsilon}, \frac{BH}{\sqrt{\varepsilon}}\right\}\log\frac{1}{\varepsilon}\right). \tag{14}$$

FastMix is used when the network is poor (ρ does not meet (10)*), the communication cost is*

$$\widetilde{\mathcal{O}}\left(\frac{1}{\sqrt{1-\rho}}\left(\tau + \max\left\{(\tau+1)(1+L)^2, \frac{B^2(1+H)}{H\varepsilon}, \frac{B}{\sqrt{\varepsilon}}\right\}\right)\log\frac{1}{\varepsilon}\right).$$

The effectness of H. The theorem does not imply that larger H is always better. Increasing H improves the transient communication complexity through the factor $H\alpha$, but it also enlarges the ultimate achievable precision. In well-connected networks, moderate increases in H can substantially reduce communication rounds. On poorly connected network, or when α is held fixed, too large H can be harmful.

Q-learning is a central application of stochastic approximation in reinforcement learning. Since the Bellman optimality operator is naturally contractive under the ℓ_∞ norm, Q-learning serves as a representative example of the general-norm setting addressed by Theorem 1.

4.2 Application in Decentralized Q-learning

We specialize the general decentralized SA framework to fixed-behavior tabular decentralized Q-learning. Let $\mathcal{S}$ and $\mathcal{A}$ be finite, fix $\gamma_{\text{disc}} \in (0,1)$, and for each agent $i \in [m]$ let $r_i : \mathcal{S}\times\mathcal{A} \to [-R_{\max}, R_{\max}]$ be a deterministic reward function and $P_i(\cdot \mid s,a)$ its local transition kernel. Each agent follows a fixed behavior policy $\pi_i(\cdot \mid s)$, independent of the current Q-iterate. The local sample stream is $Y_i^{(\ell)} := \big(S_i^{(\ell)}, A_i^{(\ell)}, R_i^{(\ell)}, S_i^{(\ell+1)}\big)$, with $R_i^{(\ell)} = r_i\big(S_i^{(\ell)}, A_i^{(\ell)}\big)$and $S_i^{(\ell+1)} \sim P_i(\cdot \mid$

$S_i^{(\ell)}, A_i^{(\ell)}$), and stationary distribution $\mu_i^{\pi_i}$. The corresponding Bellman optimality operator is in the form of

$$(T_i Q)(s,a) := r_i(s,a) + \gamma_{\text{disc}} \sum_{s' \in \mathcal{S}} P_i(s' \mid s,a) \max_{a' \in \mathcal{A}} Q(s',a'). \tag{15}$$

Let $d := |\mathcal{S}||\mathcal{A}|$ and let $e_{s,a} \in \mathbb{R}^d$ be the coordinate basis vector associated with (s,a). For a sample $y = (s,a,r,s')$, define the sampled Bellman update operator

$$F_i(Q,y) := Q + e_{s,a}\Big(r + \gamma_{\text{disc}} \max_{a' \in \mathcal{A}} Q(s',a') - Q(s,a)\Big). \tag{16}$$

With the local samples $Y_{i,t}^k := Y_i^{(kH+t)}$, the inner loop of Algorithm 1 reduces to standard tabular Q-learning updates performed locally at each agent. Let $r_{p,Q}$ denote the Markovian mixing rate induced by the behavior policy and the environment.

We now identify the correct mean operator seen by the decentralized stochastic approximation setting. Let $d_i(s,a) := P_{\mu_i^{\pi_i}}\big((S_i,A_i) = (s,a)\big)$, $\bar{d}(s,a) := \frac{1}{m}\sum_{i=1}^m d_i(s,a)$, and $d_{\min} := \min_{(s,a)\in\mathcal{S}\times\mathcal{A}} \bar{d}(s,a)$, and assume $d_{\min} > 0$. Define $\bar{D} := \text{diag}(\bar{d}(s,a) : (s,a) \in \mathcal{S}\times\mathcal{A})$, together with the coverage-weighted Bellman operator

$$(\widetilde{T}Q)(s,a) := \frac{\sum_{i=1}^m d_i(s,a)\,(T_i Q)(s,a)}{\sum_{i=1}^m d_i(s,a)}. \tag{17}$$

Appendix B shows that the averaged mean operator of the decentralized SA model has the form of $\bar{F}_Q(Q) = Q + \bar{D}(\widetilde{T}Q - Q)$ with Assumption 1 guaranteed under contraction norm $\|\cdot\|_c = \|\cdot\|_\infty$ and constants $L_Q = 1 + \gamma_{\text{disc}}$, $\beta_Q = 1 - (1-\gamma_{\text{disc}})d_{\min}$ and $B_Q = \frac{2R_{\max}}{1-\gamma_{\text{disc}}}$. Thus, the fixed point coincides with the unique fixed point $Q^\star$ of $\widetilde{T}$. We can derive the following corollary:

Corollary 2 (Decentralized Q-learning specialization)**.** *Assume the tabular Q-learning setting above, the network assumption, and geometric Markov mixing* (4) *hold. Run Algorithm 1 with Bellman update F_i defined above, constant stepsize α, and local length H.*

Suppose $\tau_Q := \widetilde{O}\left(\frac{1}{H(1-r_{p,Q})}\right)$, $\alpha := \widetilde{\mathcal{O}}\left(\frac{1}{H(\tau_Q+1)(1+L_Q)^2}\right)$ *and* $\rho_{\text{eff}} := \widetilde{\mathcal{O}}\left(\frac{1}{\sqrt{m}(\tau_Q+1)}\right)$, *let* $\mathcal{P}_Q(\alpha,H) := 1 + H + \alpha H^2(1+L_Q)^2 + \alpha^2 H^3(1+L_Q)^2$. *Then, for all* $k \geq \tau_Q$,

$$\mathbb{E}\big[R_k^2\big] \leq \widetilde{\mathcal{O}}\left(\left(1 - \frac{\alpha\kappa_Q H}{32}\right)^{k-\tau_Q}\right) + \widetilde{\mathcal{O}}\left(\alpha B_Q^2\,\mathcal{P}_Q(\alpha,H)\right), \tag{18}$$

$$\mathbb{E}\big[S_k^2\big] \leq \widetilde{\mathcal{O}}\left(\left(1 - \frac{\alpha\kappa_Q H}{32}\right)^{k-\tau_Q}\right) + \widetilde{\mathcal{O}}\left(\rho_{\text{eff}}^2\alpha^2 H^2 B_Q^2\left(1 + \alpha(1+L_Q)^2\mathcal{P}_Q(\alpha,H)\right)\right). \tag{19}$$

For ε *precision, with* $\alpha = \Theta(\varepsilon)$ *and* $\kappa_Q = 1 - \frac{1+\beta_Q}{2}\left(1 + \frac{1}{e}\left[\left(\frac{1+\beta_Q}{2\beta_Q}\right)^2 - 1\right]\right)^{-1/2}$, *the communication and sample complexities are* $\widetilde{\mathcal{O}}\left(\tau_Q + \frac{1}{\kappa_Q H\varepsilon}\log\frac{1}{\varepsilon}\right)$ *and* $\widetilde{\mathcal{O}}\left(\tau_Q H + \frac{1}{\kappa_Q\varepsilon}\log\frac{1}{\varepsilon}\right)$ *respectively.*

4.3 Analysis sketch

We briefly explain the main proof ideas behind Theorem 1. The full proof is given in Appendix A. The analysis tracks two coupled quantities: the consensus error among all agents and the estimation error of the averaged iterates. In terms of the consensus error S_k, the multiple local steps may lead to increasing, such that $S_{k,H} \leq 2S_k + \tilde{O}\left(\alpha H\left(L\max_{0\leq t\leq H-1} R_{k,t} + B\right)\right)$. The shrinking of the consensus error is brought by gossip communication after these local computations, through $S_{k+1} \leq \rho_{\text{eff}}\delta\sqrt{m}S_{k,H}$. Once ρ_{eff} is small enough, the consensus error will decrease overall. For the estimation gap, the averaged operator $\bar{F}$ is only contractive under a general norm $\|.\|_c$, which is not necessarily smooth and hard to do descent analysis. We adopt a generalized Moreau envelope [9], $M(.) : \mathbb{R}^d \to \mathbb{R}$, for smoothing the general norm and we use $\Phi_{k,t} := M(\bar{z}_t^k - x^\star)$ as potential function to characterize estimation gap's decreasing.

In the analysis, the difficulty mainly comes from controlling two types of terms: i) drift bias of multiple local steps, as there is heterogeneity between $\bar{F}_i$ and $\bar{F}$, ii) the error of Markovian sampling

when using $F_i(z_i^{k,t}, Y_i^{k,t})$ as the estimation of $\bar{F}_i(z_i^{k,t})$. The drift bias can be decomposed into the inner-loop consensus and estimation gaps, say $S_{k-1,t}^2$ and $R_{k-1,t}^2 (t \in \{0, \cdots, H-1\})$. All these inner-loop errors can be controlled by the error S_{k-1}^2 and R_{k-1}^2 from previous iteration $k-1$. In terms of bounding the sampling error in $F_i(z_i^{k,t}, Y_{i,t}^k)$, we use the methodology from [28]. Conditional on the history from τ communication rounds earlier, the current sample $Y_{i,t}^k$ evolves by at least $H\tau$ Markov transitions. By the mixing condition, its distribution is within $\widetilde{O}(\epsilon)$ of the stationary distribution μ_i. This motivates controlling the Markovian sampling bias by comparing the current iterate with the delayed iterate at round $k-\tau$ and adding a delayed error term $S_{k-\tau}^2$ (or $\Phi_{k-\tau,0}$) into the final Lyapunov function.

Finally, following recursions are obtained.

$$
\begin{aligned}
&\mathbb{E}[\Phi_{k+1,0}] \leq \left(1 - \frac{\alpha c_0 H}{8}\right) \mathbb{E}[\Phi_{k,0}] + \widetilde{O}\left(\alpha H L^2\right) \mathbb{E}[S_{k,0}^2] + \widetilde{O}\left(\alpha^2 H^2 \tau (2L+1) L\right) \mathbb{E}[\Gamma_{k,\tau}^2] + \tilde{O}(\alpha^2 H B^2), \\
&\mathbb{E}[\Gamma_{k,\tau}^2] = \widetilde{O}(\mathbb{E}[\Phi_{k-\tau,0}]) + \widetilde{O}(\mathbb{E}[S_{k-\tau,0}^2]) + \widetilde{O}(B^2), \\
&\mathbb{E}[\Phi_{k-\tau,0}] = \mathbb{E}[\Phi_{k,0}] + \widetilde{O}(\tau^2 \alpha^2 H^2) \mathbb{E}[S_{k-\tau,0}^2] + \widetilde{O}\left(\tau^2 \alpha^2 H^2 B^2\right), \\
&\mathbb{E}[S_{k+1,0}^2] = \widetilde{O}(\rho_{\text{eff}}^2 \mathbb{E}[S_{k,0}^2]) + \widetilde{O}(\rho_{\text{eff}}^2 \alpha^2 H^2 L^2) \mathbb{E}[\Phi_{k,0}^2] + \widetilde{O}(\rho_{\text{eff}}^2 \alpha^2 H^2 B^2).
\end{aligned}
\tag{20}
$$

The details are in Appendix A.4. The estimation gap and consensus error can be combined through a Lyapunov function of the form $V^k = \mathbb{E}[\Phi_{k,0}] + c_S \mathbb{E}[S_k^2] + c_S \mathbb{E}[S_{k-\tau}^2]$, where $c_S > 0$ is chosen to balance the consensus terms. Under appropriate stepsize and network conditions, it holds

$$
V^{k+1} \leq (1 - c_0 \alpha H) V^k + \widetilde{\mathcal{O}}\left(\alpha^2 H B^2 \left(1 + H + \alpha H^2 (1+L)^2 + \alpha^2 H^3 (1+L)^2\right)\right),
$$

then the error ball for estimation gap is

$$
\widetilde{\mathcal{O}}\left(\alpha B^2 \left(1 + H + \alpha H^2 (1+L)^2 + \alpha^2 H^3 (1+L)^2\right)\right).
$$

The consensus gap has error ball

$$
\widetilde{\mathcal{O}}\left(\rho_{\text{eff}}^2 \alpha^2 H^2 B^2 \left[1 + \alpha (1+L)^2 \left(1 + H + \alpha H^2 (1+L)^2 + \alpha^2 H^3 (1+L)^2\right)\right]\right).
$$

5 Experiments

In this section, we evaluate DLSA on decentralized Q-learning. The experiments are designed to verify the two implications of our theory: multiple local updates can reduce the number of communication rounds needed to reach a target estimation error, while the choice of H also affects the sample complexity and the final behavior. Following Algorithm 1, one outer round consists of H local Q-learning updates followed by one communication phase. The stepsize is constant within each run and tuned separately for different H. And For clearer visualization, we plot ℓ_2 error in all figures.

We first consider a decentralized Q-learning problem with discount factor $\gamma = 0.95$, whose Bellman optimality operator is naturally contractive under the non-Euclidean ℓ_∞ norm. We generate $m = 10$ agents, each with its own MDP, and connect them through a fixed path graph with a doubly-stochastic mixing matrix W given by Metropolis weights, i.e., $w_{ij} = 1/(1 + \max\{d_i, d_j\})$ for each edge (i, j) and $w_{ii} = 1 - \sum_{j \neq i} w_{ij}$. Each agent repeatedly samples transitions from its own MDP and maintains a local Q-table $Q_i \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$. We compute a reference solution $Q^\star$ as the fixed point of the averaged Bellman optimality operator and report the network-average table error

$$
\text{err}_k := \frac{1}{m} \sum_{i=1}^{m} \|Q_{i,k} - Q^\star\|_2.
$$

Figure 1 reports err_k against the number of communication rounds. In each communication round, every agent performs H local Q-learning updates with stepsize α_H and then executes one gossip step using W. We sweep H and compare DLSA with DCSA [38], which corresponds to the $H = 1$ case. As shown in Figure 1, DLSA with $H \geq 3$ reaches the same error level in fewer communication rounds than the $H = 1$ baseline, confirming the communication-efficiency prediction of Theorem 1.

The aggressive-stepsize ablation remains above the tuned runs, showing that poor stepsize tuning can obscure the benefit of local updates.

Figure 2 evaluates the sample complexity in the same decentralized Q-learning setting. We plot the same error err_k against the number of samples per agent for $H \in \{1, 3, 6, 9, 12\}$. The curves show that moderate local-update lengths such as $H = 3$ and $H = 6$ remain competitive in terms of samples per agent, while larger choices such as $H = 9$ and $H = 12$ converge more slowly. This agrees with Corollary 2: increasing H can reduce communication cost, but too large H may increase sample complexity.

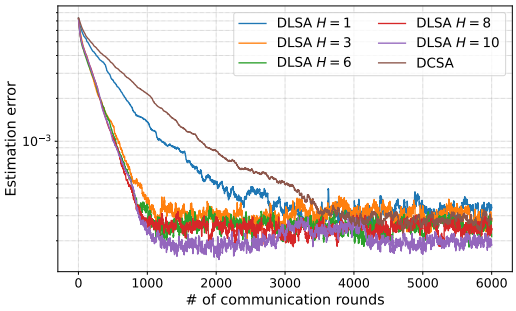


Figure 1: Estimation error analysis

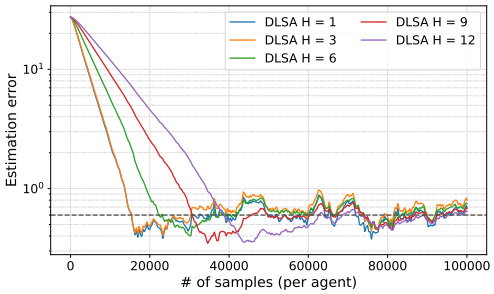


Figure 2: Sample complexity analysis

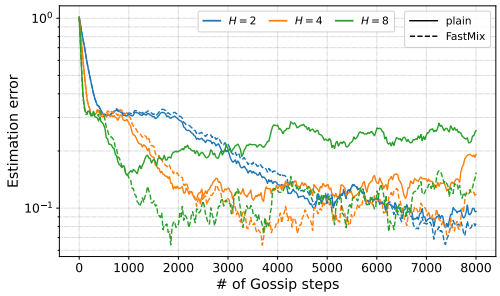


Figure 3: FastMix comparison

We also test DLSA-FastMix on a poorly-connected path graph with $m = 8$ agents over heterogeneous random MDPs, where $\gamma = 0.9$ and $H \in \{2, 4, 8\}$. We compare DLSA-FastMix with $n_\rho = 12$ polynomial-filter gossip steps per round against plain DLSA with one gossip step per round, and plot the per-agent estimation error against the total number of gossip steps. The step sizes are tuned separately for different values of H, and for each fixed H, DLSA and DLSA-FastMix use the same stepsize. As shown in Figure 3, FastMix reaches a lower steady-state error than plain gossip for every H, confirming that the accelerated multi-consensus routine improves the effective mixing on poorly-connected graphs. Moreover, the gap between FastMix and plain gossip becomes larger as H increases, which is consistent with Theorem 1: larger H amplifies the effect of the consensus error, and thus a smaller effective network mixing factor ρ_{eff} becomes more important.

Additional experiments on linear stochastic approximation and stochastic gradient descent with Markovian noise are reported in Appendix C.

6 Conclusion

In this work, we proposed DLSA, a communication-efficient decentralized stochastic approximation method that performs multiple local SA updates between successive communication steps, thereby substantially reducing the number of communication rounds required to approach the desired fixed point, and further enabling additional gains when combined with communication acceleration techniques. On the theoretical side, we established a finite-time convergence guarantee for DLSA under the assumption that the averaged operator is contractive under an arbitrary norm, which allows our analysis to handle non-Euclidean contractions, including $\|\cdot\|_\infty$-type settings arising in tabular Q-learning. Finally, we conducted extensive numerical experiments on reinforcement learning tasks, linear stochastic approximation and Markovian stochastic gradient descent, empirically validating the effectiveness of DLSA and demonstrating significant communication savings over existing decentralized SA baselines while maintaining strong practical performance.

Despite its advantages, our theory suggests that using a larger number of local updates H may amplify the resulting error ball, and hence requires additional stepsize tuning to ensure the final accuracy meets a target precision. A promising future direction is to develop adaptive stepsize strategies that achieve communication savings without requiring prior knowledge of problem-dependent constants, while preserving strong convergence guarantees.

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A Detailed Analysis of DLSA

This section completes the convergence analysis of DLSA.

A.1 Preliminaries

Moreau envelope. We introduce the generalized Moreau envelope $M(\cdot) : \mathbb{R}^d \to \mathbb{R}$, with parameter $\theta > 0$ and auxiliary norm $\|\cdot\|_s$,

$$M(x) := \min_{u\in\mathbb{R}^d} \left\{ \frac{1}{2}\|u\|_c^2 + \frac{1}{2\theta}\|u - x\|_s^2 \right\}.$$

Then, there exist constants $\ell_{cm}, u_{cm}, \ell_{cs} > 0$, such that

$$\ell_{cm}\|x\|_M \le \|x\|_c \le u_{cm}\|x\|_M, \quad \ell_{cs}\|x\|_s \le \|x\|_c, \qquad \forall x \in \mathbb{R}^d.$$

Moreover, $M(\cdot)$ is differentiable and L_{ms}-smooth w.r.t. $\|\cdot\|_s$, i.e., for all $x, y \in \mathbb{R}^d$,

$$M(y) \le M(x) + \langle \nabla M(x), y - x\rangle + \frac{L_{ms}}{2}\|y - x\|_s^2. \tag{21}$$

We also use the equivalent gradient Lipschitz property

$$\|\nabla M(x) - \nabla M(y)\|_s^\star \le L_{ms}\|x - y\|_s, \qquad \forall x, y \in \mathbb{R}^d. \tag{22}$$

There exists a norm $\|\cdot\|_M$, such that for any $x \in \mathbb{R}^d$,

$$M(x) = \frac{1}{2}\|x\|_M^2, \qquad \|\nabla M(x)\|_M^\star = \|x\|_M, \qquad \langle \nabla M(x), x\rangle = \|x\|_M^2 = 2M(x). \tag{23}$$

In terms of the selection of the auxiliary norm $\|.\|_s$ given the β-contraction of $\bar{F}$ w.r.t. $\|.\|_c$, we assume the auxiliary norm $\|.\|_s$ is chosen well, such that

$$\kappa := 1 - \beta\frac{u_{cm}}{\ell_{cm}} \in (0, 1).$$

Markov-chain indexing and filtrations. For each agent i, the samples $\{Y_{i,t}^k\}_{k\ge 0, 0\le t\le H-1}$ form a single trajectory of a time-homogeneous Markov chain (with stationary distribution μ_i as assumed in the main text). It is sometimes convenient to flatten the double index via the global time $n := kH + t$ and write $Y_{i,t}^k = \widetilde{Y}_i^n$. We denote by $\mathcal{F}_{k,t}$ the sigma-field generated by all iterates and samples up to the beginning of inner step (k, t), i.e.,

$$\mathcal{F}_{k,t} := \sigma\Big(\{z_{i,s}^\ell, Y_{i,s}^\ell : 0 \le \ell < k, 0 \le s \le H-1, i \in [m]\} \cup \{z_{i,s}^k, Y_{i,s}^k : 0 \le s < t, i \in [m]\}\Big),$$

so that $z_{i,t}^k$ is $\mathcal{F}_{k,t}$-measurable and $Y_{i,t}^k$ is the next Markov state revealed at step (k, t).

A.2 Consensus error for Algorithm 1

Denote $J := \frac{1}{m}\mathbf{1}\mathbf{1}^\top$, then $\rho = \|W - J\|_2 \in [0, 1)$. The communication step can be written compactly as $X^{k+1} = (W \otimes I_d)Z_H^k$, and the post-communication disagreement satisfies $X^{k+1} - (\mathbf{1} \otimes \bar{x}^{k+1}) = ((W - J) \otimes I_d)\left(Z_H^k - (\mathbf{1} \otimes \bar{z}_H^k)\right)$. Although the analysis of consensus error is for one step of gossip communication with shrinking factor ρ, it can be directly replaced with ρ_{fm} if FastMix is adopted when the network is poorly connected, and the analysis keeps nearly unchanged.

This is the starting point for deriving contraction of consensus error.

Lemma 1 (Inner-loop drift bounds)**.** *Fix an outer round $k \ge 0$ and consider the inner iterates $\{z_{i,t}^k\}_{t=0}^H$ generated by Algorithm 1. Then for every $t = 0, 1, \ldots, H-1$, the following inequalities hold (pathwise, i.e., for any realization of the Markov chains):*

$$R_{k,t+1} \le (1 - \alpha + \alpha L)\, R_{k,t} + \alpha L\, S_{k,t} + \alpha B, \tag{24}$$

$$S_{k,t+1} \le (1 - \alpha + 2\alpha L)\, S_{k,t} + 2\alpha\big(LR_{k,t} + B\big). \tag{25}$$

Moreover, letting

$$b := 1 - \alpha + \alpha L, \qquad a := 1 - \alpha + 2\alpha L,$$

we have the explicit unrolled forms for all $t = 1, 2, \ldots, H$*:*

$$R_{k,t} \le b^t R_{k,0} + \alpha L \sum_{s=0}^{t-1} b^{\,t-1-s} S_{k,s} + \alpha B \sum_{s=0}^{t-1} b^{\,t-1-s}, \tag{26}$$

$$S_{k,t} \le a^t S_{k,0} + 2\alpha \sum_{s=0}^{t-1} a^{\,t-1-s} \big(L R_{k,s} + B \big). \tag{27}$$

In particular, if $\alpha L H \le \frac{1}{4}$*, then*

$$S_{k,H} \le 2 S_{k,0} + 4\alpha H \Big(L \max_{0 \le s \le H-1} R_{k,s} + B \Big). \tag{28}$$

Proof. From the inner update $z^k_{i,t+1} = (1-\alpha) z^k_{i,t} + \alpha F_i(z^k_{i,t}, Y^k_{i,t})$ and the definition $\bar{z}^k_t = \frac{1}{m}\sum_{i=1}^m z^k_{i,t}$, we have

$$\bar{z}^k_{t+1} = (1-\alpha)\bar{z}^k_t + \alpha \frac{1}{m} \sum_{i=1}^m F_i(z^k_{i,t}, Y^k_{i,t}). \tag{29}$$

Hence, by triangle inequality,

$$\begin{aligned} R_{k,t+1} &= \|\bar{z}^k_{t+1} - x^\star\|_c \\ &= \Big\| (1-\alpha)(\bar{z}^k_t - x^\star) + \alpha \big(\frac{1}{m} \sum_{i=1}^m F_i(z^k_{i,t}, Y^k_{i,t}) - x^\star \big) \Big\|_c \\ &\le (1-\alpha) \|\bar{z}^k_t - x^\star\|_c + \alpha \Big\| \frac{1}{m} \sum_{i=1}^m F_i(z^k_{i,t}, Y^k_{i,t}) - x^\star \Big\|_c \\ &\le (1-\alpha) R_{k,t} + \alpha \frac{1}{m} \sum_{i=1}^m \|F_i(z^k_{i,t}, Y^k_{i,t}) - x^\star\|_c. \end{aligned}$$

For each i, insert and subtract $F_i(x^\star, Y^k_{i,t})$ and use Assumption 1(i),(iii):

$$\begin{aligned} \|F_i(z^k_{i,t}, Y^k_{i,t}) - x^\star\|_c &\le \|F_i(z^k_{i,t}, Y^k_{i,t}) - F_i(x^\star, Y^k_{i,t})\|_c + \|F_i(x^\star, Y^k_{i,t}) - x^\star\|_c \\ &\le L \|z^k_{i,t} - x^\star\|_c + B. \end{aligned}$$

Also,

$$\|z^k_{i,t} - x^\star\|_c \le \|z^k_{i,t} - \bar{z}^k_t\|_c + \|\bar{z}^k_t - x^\star\|_c = \|z^k_{i,t} - \bar{z}^k_t\|_c + R_{k,t}.$$

Averaging over i gives

$$\frac{1}{m} \sum_{i=1}^m \|z^k_{i,t} - x^\star\|_c \le S_{k,t} + R_{k,t}.$$

Combining the last three displays yields

$$\begin{aligned} R_{k,t+1} &\le (1-\alpha) R_{k,t} + \alpha \Big(L(S_{k,t} + R_{k,t}) + B \Big) \\ &= (1 - \alpha + \alpha L) R_{k,t} + \alpha L S_{k,t} + \alpha B, \end{aligned}$$

which is (24).

From the inner update and (29),

$$z^k_{i,t+1} - \bar{z}^k_{t+1} = (1-\alpha)(z^k_{i,t} - \bar{z}^k_t) + \alpha \Big(F_i(z^k_{i,t}, Y^k_{i,t}) - \frac{1}{m} \sum_{i=1}^m F_j(z^k_{j,t}, Y^k_{j,t}) \Big).$$

Taking $\|\cdot\|_c$ and applying triangle inequality,

$$\|z^k_{i,t+1} - \bar{z}^k_{t+1}\|_c \le (1-\alpha) \|z^k_{i,t} - \bar{z}^k_t\|_c + \alpha \Big\| F_i(z^k_{i,t}, Y^k_{i,t}) - \frac{1}{m} \sum_{i=1}^m F_j(z^k_{j,t}, Y^k_{j,t}) \Big\|_c. \tag{30}$$

We further split the last term by adding and subtracting $F_i(\bar{z}_t^k, Y_{i,t}^k)$ and $\frac{1}{m}\sum_{i=1}^m F_j(\bar{z}_t^k, Y_{j,t}^k)$:

$$\Big\|F_i(z_{i,t}^k, Y_{i,t}^k) - \frac{1}{m}\sum_{i=1}^m F_j(z_{j,t}^k, Y_{j,t}^k)\Big\|_c \le \underbrace{\|F_i(z_{i,t}^k, Y_{i,t}^k) - F_i(\bar{z}_t^k, Y_{i,t}^k)\|_c}_{\text{(A)}}$$
$$+ \underbrace{\Big\|F_i(\bar{z}_t^k, Y_{i,t}^k) - \frac{1}{m}\sum_{i=1}^m F_j(\bar{z}_t^k, Y_{j,t}^k)\Big\|_c}_{\text{(B)}} + \underbrace{\Big\|\frac{1}{m}\sum_{i=1}^m \big(F_j(\bar{z}_t^k, Y_{j,t}^k) - F_j(z_{j,t}^k, Y_{j,t}^k)\big)\Big\|_c}_{\text{(C)}}. \tag{31}$$

We bound (A), (B), (C) separately.

For (A), by Assumption 1 (i),

$$\text{(A)} \le L\|z_{i,t}^k - \bar{z}_t^k\|_c.$$

For (B), using $\|u - \frac{1}{m}\sum_{i=1}^m v_j\|_c \le \|u\|_c + \|\frac{1}{m}\sum_{i=1}^m v_j\|_c \le \|u\|_c + \frac{1}{m}\sum_{i=1}^m \|v_j\|_c$, we obtain

$$\text{(B)} \le \|F_i(\bar{z}_t^k, Y_{i,t}^k)\|_c + \frac{1}{m}\sum_{i=1}^m \|F_j(\bar{z}_t^k, Y_{j,t}^k)\|_c.$$

For any i, by Assumption 1 (i), (iii),

$$\begin{aligned}\|F_i(\bar{z}_t^k, Y_{i,t}^k)\|_c &\le \|F_i(\bar{z}_t^k, Y_{i,t}^k) - F_i(x^\star, Y_{i,t}^k)\|_c + \|F_i(x^\star, Y_{i,t}^k)\|_c \\ &\le L\|\bar{z}_t^k - x^\star\|_c + B = LR_{k,t} + B.\end{aligned}$$

Hence,

$$\text{(B)} \le 2(LR_{k,t} + B).$$

For (C), by Jensen and Assumption 1 (i),

$$\text{(C)} \le \frac{1}{m}\sum_{i=1}^m \|F_i(\bar{z}_t^k, Y_{i,t}^k) - F_i(z_{i,t}^k, Y_{i,t}^k)\|_c \le \frac{1}{m}\sum_{i=1}^m L\|\bar{z}_t^k - z_{i,t}^k\|_c = LS_{k,t}.$$

Plugging the bounds for (A)–(C) into (31) and then into (30) yields

$$\|z_{i,t+1}^k - \bar{z}_{t+1}^k\|_c \le (1-\alpha)\|z_{i,t}^k - \bar{z}_t^k\|_c + \alpha\Big(L\|z_{i,t}^k - \bar{z}_t^k\|_c + 2(LR_{k,t} + B) + LS_{k,t}\Big).$$

Rearranging,

$$\|z_{i,t+1}^k - \bar{z}_{t+1}^k\|_c \le (1 - \alpha + \alpha L)\|z_{i,t}^k - \bar{z}_t^k\|_c + \alpha L S_{k,t} + 2\alpha(LR_{k,t} + B).$$

Averaging this inequality over $i \in [m]$ gives

$$\begin{aligned}S_{k,t+1} &= \frac{1}{m}\sum_{i=1}^m \|z_{i,t+1}^k - \bar{z}_{t+1}^k\|_c \\ &\le (1 - \alpha + \alpha L)\frac{1}{m}\sum_{i=1}^m \|z_{i,t}^k - \bar{z}_t^k\|_c + \alpha L S_{k,t} + 2\alpha(LR_{k,t} + B) \\ &= (1 - \alpha + 2\alpha L)S_{k,t} + 2\alpha(LR_{k,t} + B),\end{aligned}$$

which is (25).

We first show (27). Starting from (25), for $t \ge 0$,

$$S_{k,t+1} \le aS_{k,t} + 2\alpha(LR_{k,t} + B).$$

Apply this inequality recursively:

$$\begin{aligned}&S_{k,1} \le aS_{k,0} + 2\alpha(LR_{k,0} + B), \\ &S_{k,2} \le aS_{k,1} + 2\alpha(LR_{k,1} + B) \le a^2 S_{k,0} + 2\alpha\big(a(LR_{k,0} + B) + (LR_{k,1} + B)\big),\end{aligned}$$

and by induction one obtains

$$S_{k,t} \le a^t S_{k,0} + 2\alpha \sum_{s=0}^{t-1} a^{\,t-1-s}(LR_{k,s} + B),$$

which is (27). The bound (28) follows by using $a \le 1 + 2\alpha L$ and, for $t = H$,

$$\sum_{s=0}^{H-1} a^{\,H-1-s} \le H \max_{0 \le r \le H-1} a^r \le H a^{H-1} \le H a^H \le 2H,$$

together with $\max_{0 \le s \le H-1}(LR_{k,s} + B)$ pulled out of the sum in (27).

For every $t \ge 0$ we have

$$R_{k,t+1} \le bR_{k,t} + \alpha L S_{k,t} + \alpha B, \qquad \text{where } b := 1 - \alpha + \alpha L. \tag{32}$$

We now unroll (32) explicitly. First,

$$R_{k,1} \le bR_{k,0} + \alpha L S_{k,0} + \alpha B.$$

Applying (32) again with $t = 1$ yields

$$\begin{aligned} R_{k,2} &\le bR_{k,1} + \alpha L S_{k,1} + \alpha B \\ &\le b\Big(bR_{k,0} + \alpha L S_{k,0} + \alpha B\Big) + \alpha L S_{k,1} + \alpha B \\ &= b^2 R_{k,0} + \alpha L\Big(bS_{k,0} + S_{k,1}\Big) + \alpha B\Big(b + 1\Big). \end{aligned}$$

Similarly, applying (32) once more gives

$$\begin{aligned} R_{k,3} &\le bR_{k,2} + \alpha L S_{k,2} + \alpha B \\ &\le b\Big(b^2 R_{k,0} + \alpha L(bS_{k,0} + S_{k,1}) + \alpha B(b+1)\Big) + \alpha L S_{k,2} + \alpha B \\ &= b^3 R_{k,0} + \alpha L\Big(b^2 S_{k,0} + bS_{k,1} + S_{k,2}\Big) + \alpha B\Big(b^2 + b + 1\Big). \end{aligned}$$

From these first few cases the pattern is clear. We prove by induction that for any integer $t \ge 1$

$$R_{k,t} \le b^t R_{k,0} + \alpha L \sum_{s=0}^{t-1} b^{\,t-1-s} S_{k,s} + \alpha B \sum_{s=0}^{t-1} b^{\,t-1-s}. \tag{33}$$

For the base case $t = 1$, (33) is exactly $R_{k,1} \le bR_{k,0} + \alpha L S_{k,0} + \alpha B$. Assume (33) holds for some $t \ge 1$. Then by (32),

$$\begin{aligned} R_{k,t+1} &\le bR_{k,t} + \alpha L S_{k,t} + \alpha B \\ &\le b\Big(b^t R_{k,0} + \alpha L \sum_{s=0}^{t-1} b^{\,t-1-s} S_{k,s} + \alpha B \sum_{s=0}^{t-1} b^{\,t-1-s}\Big) + \alpha L S_{k,t} + \alpha B \\ &= b^{t+1} R_{k,0} + \alpha L \sum_{s=0}^{t-1} b^{\,t-s} S_{k,s} + \alpha B \sum_{s=0}^{t-1} b^{\,t-s} + \alpha L S_{k,t} + \alpha B \\ &= b^{t+1} R_{k,0} + \alpha L \sum_{s=0}^{t} b^{\,t-s} S_{k,s} + \alpha B \sum_{s=0}^{t} b^{\,t-s}, \end{aligned}$$

which is exactly (33) with t replaced by $t + 1$. This completes the induction and proves (26). □

Lemma 2 (Consensus error after one communication)**.** *Fix $k \ge 0$ and let $\{z_{i,H}^k\}_{i=1}^m$ be the iterates of H local steps from Algorithm 1, and let $\{x_i^{k+1}\}_{i=1}^m$ be the iterates after local communications,*

$$x_i^{k+1} = \sum_{j=1}^m w_{ij} z_{j,H}^k, \qquad i \in [m].$$

Then the one-communication step satisfies

$$S_{k+1} \le \rho \, \frac{u_{2c}}{\ell_{2c}} \sqrt{m} \, S_{k,H}, \tag{34}$$

where $\ell_{2c}, u_{2c} > 0$ are norm equivalence parameters, such that

$$\ell_{2c}\|v\|_2 \le \|v\|_c \le u_{2c}\|v\|_2, \quad \forall v \in \mathbb{R}^d. \tag{35}$$

In particular, if $\alpha LH \le \frac{1}{4}$, then Lemma 1 implies

$$S_{k+1} \le \rho \, \frac{u_{2c}}{\ell_{2c}} \sqrt{m} \left(2S_k + 4\alpha H \big(L \max_{0 \le s \le H-1} R_{k,s} + B \big) \right). \tag{36}$$

Proof. By the property of mixing matrix W, we have

$$\left(\frac{1}{m} \sum_{i=1}^m \|x_i^{k+1} - \bar{x}^{k+1}\|_2^2 \right)^{1/2} \le \rho \left(\frac{1}{m} \sum_{i=1}^m \|z_{i,H}^k - \bar{z}_H^k\|_2^2 \right)^{1/2}. \tag{37}$$

We bound S_{k+1} from above using (35),

$$S_{k+1} = \frac{1}{m} \sum_{i=1}^m \|x_i^{k+1} - \bar{x}^{k+1}\|_c \le \frac{u_{2c}}{m} \sum_{i=1}^m \|x_i^{k+1} - \bar{x}^{k+1}\|_2 \le u_{2c} \left(\frac{1}{m} \sum_{i=1}^m \|x_i^{k+1} - \bar{x}^{k+1}\|_2^2 \right)^{1/2}. \tag{38}$$

On the other hand,

$$\left(\frac{1}{m} \sum_{i=1}^m \|z_{i,H}^k - \bar{z}_H^k\|_2^2 \right)^{1/2} \le \frac{1}{\ell_{2c}} \left(\frac{1}{m} \sum_{i=1}^m \|z_{i,H}^k - \bar{z}_H^k\|_c^2 \right)^{1/2} \le \frac{\sqrt{m}}{\ell_{2c}} \, S_{k,H}. \tag{39}$$

Then

$$S_{k+1} \le u_{2c} \left(\frac{1}{m} \sum_{i=1}^m \|x_i^{k+1} - \bar{x}^{k+1}\|_2^2 \right)^{1/2} \le u_{2c}\rho \left(\frac{1}{m} \sum_{i=1}^m \|z_{i,H}^k - \bar{z}_H^k\|_2^2 \right)^{1/2} \le \rho \, \frac{u_{2c}}{\ell_{2c}} \sqrt{m} \, S_{k,H}.$$

If $\alpha LH \le \frac{1}{4}$, Lemma 1 gives $S_{k,H} \le 2S_k + 4\alpha H(L \max_{0 \le s \le H-1} R_{k,s} + B)$. Substituting this into (34) yields (36). □

A.3 Estimation gap along the inner-loop average

Define the potential

$$\Phi_{k,t} := M(\bar{z}_t^k - x^\star), \qquad t = 0, 1, \ldots, H.$$

Then $\Phi_{k,0} = M(\bar{x}^k - x^\star)$ and $\Phi_{k,H} = M(\bar{x}^{k+1} - x^\star)$.

Denote $\Delta_{k,t} := \bar{z}_{t+1}^k - \bar{z}_t^k$, then by local update

$$z_{i,t+1}^k = z_{i,t}^k + \alpha(F_i(z_{i,t}^k, Y_{i,t}^k) - z_{i,t}^k)$$

we have

$$\Delta_{k,t} = \alpha \left(\frac{1}{m} \sum_{i=1}^m F_i(z_{i,t}^k, Y_{i,t}^k) - \bar{z}_t^k \right) = \alpha \big(\bar{F}(\bar{z}_t^k) - \bar{z}_t^k \big) + \alpha \, \mathsf{E}_{k,t}^{\mathrm{cons}} + \alpha \, \mathsf{E}_{k,t}^{\mathrm{mk}}, \tag{40}$$

where

$$\mathsf{E}_{k,t}^{\mathrm{cons}} := \frac{1}{m} \sum_{i=1}^m \Big(F_i(z_{i,t}^k, Y_{i,t}^k) - F_i(\bar{z}_t^k, Y_{i,t}^k) \Big), \quad \mathsf{E}_{k,t}^{\mathrm{mk}} := \frac{1}{m} \sum_{i=1}^m \Big(F_i(\bar{z}_t^k, Y_{i,t}^k) - \bar{F}_i(\bar{z}_t^k) \Big). \tag{41}$$

In addition, we denote $\mathcal{T}_{k,t}^{\mathrm{mk}} := \alpha \, \mathbb{E}[\langle \nabla M(\bar{z}_t^k - x^\star), \mathsf{E}_{k,t}^{\mathrm{mk}} \rangle]$, a quantity relevant to Markovian noise.

Lemma 3 (Inner-step descent inequality)**.** *Fix $k \geq 0$ and $t \in \{0, 1, \ldots, H-1\}$, then*

$$\mathbb{E}[\Phi_{k,t+1}] \leq \Big(1 - \alpha\kappa + \frac{3L_{ms}\alpha^2(L+1)^2u_{cm}^2}{\ell_{cs}^2}\Big)\,\mathbb{E}[\Phi_{k,t}] + \mathcal{T}_{k,t}^{\mathrm{mk}} + \Big(\frac{\alpha L^2 u_{cm}^2}{2\kappa\,\ell_{cm}^4} + \frac{3L_{ms}\alpha^2L^2}{2\ell_{cs}^2}\Big)\,\mathbb{E}[S_{k,t}^2] + \frac{3L_{ms}\alpha^2}{2\ell_{cs}^2}\,B^2. \tag{42}$$

Proof. Applying (21) with $x = \bar{z}_t^k - x^\star$ and $y = \bar{z}_{t+1}^k - x^\star$ gives

$$\Phi_{k,t+1} \leq \Phi_{k,t} + \big\langle \nabla M(\bar{z}_t^k - x^\star), \Delta_{k,t}\big\rangle + \frac{L_{ms}}{2}\|\Delta_{k,t}\|_s^2. \tag{43}$$

For $\big\langle \nabla M(\bar{z}_t^k - x^\star), \Delta_{k,t}\big\rangle$, we separate it into three parts using (40),

$$\begin{aligned}&\big\langle \nabla M(\bar{z}_t^k - x^\star), \Delta_{k,t}\big\rangle \\ &= \big\langle \nabla M(\bar{z}_t^k - x^\star), \alpha(\bar{F}(\bar{z}_t^k) - \bar{z}_t^k)\big\rangle + \big\langle \nabla M(\bar{z}_t^k - x^\star), \alpha\mathsf{E}_{k,t}^{\mathrm{cons}}\big\rangle + \big\langle \nabla M(\bar{z}_t^k - x^\star), \alpha\mathsf{E}_{k,t}^{\mathrm{mk}}\big\rangle.\end{aligned}$$

Then, we need to bound the terms $\big\langle \nabla M(\bar{z}_t^k - x^\star), \alpha(\bar{F}(\bar{z}_t^k) - \bar{z}_t^k)\big\rangle$, $\big\langle \nabla M(\bar{z}_t^k - x^\star), \alpha\mathsf{E}_{k,t}^{\mathrm{cons}}\big\rangle$ and $\|\Delta_{k,t}\|_s^2$. The term $\big\langle \nabla M(\bar{z}_t^k - x^\star), \alpha\mathsf{E}_{k,t}^{\mathrm{mk}}\big\rangle$ is exactly $\mathcal{T}_{k,t}^{\mathrm{mk}}$ after taking expectation.

Bound $\big\langle \nabla M(\bar{z}_t^k - x^\star), \alpha(\bar{F}(\bar{z}_t^k) - \bar{z}_t^k)\big\rangle$:

Let $x_{k,t} = \bar{z}_t^k - x^\star$ as in the statement. Using $\bar{F}(x^\star) = x^\star$,

$$\bar{F}(\bar{z}_t^k) - \bar{z}_t^k = \big(\bar{F}(\bar{z}_t^k) - \bar{F}(x^\star)\big) - (\bar{z}_t^k - x^\star).$$

Therefore, by (23),

$$\begin{aligned}\big\langle \nabla M(\bar{z}_t^k - x^\star), \bar{F}(\bar{z}_t^k) - \bar{z}_t^k\big\rangle &= -\langle \nabla M(\bar{z}_t^k - x^\star), \bar{z}_t^k - x^\star\rangle + \big\langle \nabla M(\bar{z}_t^k - x^\star), \bar{F}(\bar{z}_t^k) - \bar{F}(x^\star)\big\rangle \\ &= -2M(\bar{z}_t^k - x^\star) + \big\langle \nabla M(\bar{z}_t^k - x^\star), \bar{F}(\bar{z}_t^k) - x^\star\big\rangle. \end{aligned}\tag{44}$$

Bound the second term in (44) using

$$\big\langle \nabla M(\bar{z}_t^k - x^\star), \bar{F}(\bar{z}_t^k) - x^\star\big\rangle \leq \|\nabla M(\bar{z}_t^k - x^\star)\|_M^\star \|\bar{F}(\bar{z}_t^k) - x^\star\|_M.$$

By (23), $\|\nabla M(\bar{z}_t^k - x^\star)\|_M^\star = \|\bar{z}_t^k - x^\star\|_M$. Next, by β-contraction of $\bar{F}$ in $\|\cdot\|_c$,

$$\|\bar{F}(\bar{z}_t^k) - x^\star\|_M \leq \frac{1}{\ell_{cm}}\|\bar{F}(\bar{z}_t^k) - x^\star\|_c \leq \frac{\beta}{\ell_{cm}}\|\bar{z}_t^k - x^\star\|_c \leq \frac{\beta u_{cm}}{\ell_{cm}}\|\bar{z}_t^k - x^\star\|_M.$$

Combining the last two displays yields

$$\big\langle \nabla M(\bar{z}_t^k - x^\star), \bar{F}(\bar{z}_t^k) - x^\star\big\rangle \leq \frac{\beta u_{cm}}{\ell_{cm}}\|\bar{z}_t^k - x^\star\|_M^2 = \frac{2\beta u_{cm}}{\ell_{cm}}\,M(\bar{z}_t^k - x^\star).$$

Plugging this into (44) gives

$$\big\langle \nabla M(\bar{z}_t^k - x^\star), \bar{F}(\bar{z}_t^k) - \bar{z}_t^k\big\rangle \leq -2\Big(1 - \beta\frac{u_{cm}}{\ell_{cm}}\Big)M(\bar{z}_t^k - x^\star) = -2\kappa\,\Phi_{k,t}. \tag{45}$$

Bound $\big\langle \nabla M(\bar{z}_t^k - x^\star), \alpha\mathsf{E}_{k,t}^{\mathrm{cons}}\big\rangle$:

Since

$$\|\nabla M(\bar{z}_t^k - x^\star)\|_M^\star = \|\bar{z}_t^k - x^\star\|_M \leq \frac{1}{\ell_{cm}}\|\bar{z}_t^k - x^\star\|_c = \frac{1}{\ell_{cm}}R_{k,t}. \tag{46}$$

$$\|\mathsf{E}_{k,t}^{\mathrm{cons}}\|_M \leq \frac{1}{\ell_{cm}}\|\mathsf{E}_{k,t}^{\mathrm{cons}}\|_c \leq \frac{1}{\ell_{cm}}\cdot\frac{1}{m}\sum_{i=1}^m \|F_i(z_{i,t}^k, Y_{i,t}^k) - F_i(\bar{z}_t^k, Y_{i,t}^k)\|_c \leq \frac{L}{\ell_{cm}}\,S_{k,t},$$

we have

$$\big\langle \nabla M(\bar{z}_t^k - x^\star), \mathsf{E}_{k,t}^{\mathrm{cons}}\big\rangle \leq \|\nabla M(\bar{z}_t^k - x^\star)\|_M^\star \|\mathsf{E}_{k,t}^{\mathrm{cons}}\|_M \leq \frac{L}{\ell_{cm}^2}\,R_{k,t}S_{k,t}. \tag{47}$$

We now absorb the resulting $\Phi_{k,t}$ contribution into the contractive term (45). We have $R_{k,t} = \|\bar{z}_t^k - x^\star\|_c \le u_{cm}\|\bar{z}_t^k - x^\star\|_M = u_{cm}\sqrt{2\Phi_{k,t}}$. Thus

$$\frac{L}{\ell_{cm}^2} R_{k,t} S_{k,t} \le \frac{L u_{cm}}{\ell_{cm}^2}\,\sqrt{2\Phi_{k,t}}\, S_{k,t} \le \kappa\,\Phi_{k,t} + \frac{L^2 u_{cm}^2}{2\kappa\,\ell_{cm}^4}\, S_{k,t}^2.$$

Therefore,

$$\alpha\,\langle \nabla M(\bar{z}_t^k - x^\star), \mathsf{E}_{k,t}^{\mathrm{cons}}\rangle \le \alpha\kappa\,\Phi_{k,t} + \frac{\alpha L^2 u_{cm}^2}{2\kappa\,\ell_{cm}^4}\, S_{k,t}^2. \tag{48}$$

Bound $\|\Delta_{k,t}\|_s^2$:

Using (40),

$$\|\Delta_{k,t}\|_s^2 \le \frac{\alpha^2}{\ell_{cs}^2}\left\|\frac{1}{m}\sum_{i=1}^m F_i(z_{i,t}^k, Y_{i,t}^k) - \bar{z}_t^k\right\|_c^2. \tag{49}$$

Insert and subtract $F_i(x^\star, Y_{i,t}^k)$ and $x^\star$:

$$\frac{1}{m}\sum_{i=1}^m F_i(z_{i,t}^k, Y_{i,t}^k) - \bar{z}_t^k = \frac{1}{m}\sum_{i=1}^m \big(F_i(z_{i,t}^k, Y_{i,t}^k) - F_i(x^\star, Y_{i,t}^k)\big) + \frac{1}{m}\sum_{i=1}^m \big(F_i(x^\star, Y_{i,t}^k) - x^\star\big) + (x^\star - \bar{z}_t^k).$$

By triangle inequality and Assumption 1(i),(iii),

$$\begin{aligned}\|\frac{1}{m}\sum_{i=1}^m F_i(z_{i,t}^k, Y_{i,t}^k) - \bar{z}_t^k\|_c &\le \frac{1}{m}\sum_{i=1}^m L\|z_{i,t}^k - x^\star\|_c + B + \|\bar{z}_t^k - x^\star\|_c \\ &\le L\Big(\frac{1}{m}\sum_{i=1}^m \|z_{i,t}^k - \bar{z}_t^k\|_c + \|\bar{z}_t^k - x^\star\|_c\Big) + B + R_{k,t} = (L+1)R_{k,t} + LS_{k,t} + B.\end{aligned}$$

Then

$$\|\frac{1}{m}\sum_{i=1}^m F_i(z_{i,t}^k, Y_{i,t}^k) - \bar{z}_t^k\|_c^2 \le 3\Big((L+1)^2 R_{k,t}^2 + L^2 S_{k,t}^2 + B^2\Big).$$

Plug into (49) and use $R_{k,t}^2 = \|\bar{z}_t^k - x^\star\|_c^2 \le u_{cm}^2\|\bar{z}_t^k - x^\star\|_M^2 = 2u_{cm}^2\Phi_{k,t}$,

$$\frac{L_{ms}}{2}\|\Delta_{k,t}\|_s^2 \le \frac{3L_{ms}\alpha^2(L+1)^2 u_{cm}^2}{\ell_{cs}^2}\,\Phi_{k,t} + \frac{3L_{ms}\alpha^2 L^2}{2\ell_{cs}^2}\, S_{k,t}^2 + \frac{3L_{ms}\alpha^2}{2\ell_{cs}^2}\, B^2. \tag{50}$$

Combine (45), (48), (50) and take expectation, then yield exactly (42). □

We now bound $\mathcal{T}_{k,t}^{\mathrm{mk}}$ under the geometric mixing assumption specified in (4). Given $\epsilon > 0$, we define a round-level Markovian delay

$$\tau := \Big\lceil \frac{1}{H(1-r_p)} \log\frac{c_p}{\epsilon}\Big\rceil. \tag{51}$$

For any $k \ge \tau$, $t \in \{0, \dots, H-1\}$, $i \in [m]$, the chain has evolved for at least $H\tau$ steps since $\mathcal{F}_{k-\tau,0}$, by (4),

$$d_{\mathrm{TV}}\big(\mathcal{L}(Y_{i,t}^k \mid \mathcal{F}_{k-\tau,0}), \mu_i\big) \le c_p r_p^{H\tau} \le \epsilon. \tag{52}$$

We now perform a four-term decomposition as follows:

$$\begin{aligned}\mathcal{T}_{k,t}^{\mathrm{mk}} = \frac{\alpha}{m}\sum_{i=1}^m \mathbb{E}[\Big\langle \nabla M(\bar{z}_t^k - x^\star) - \nabla M(\bar{x}^{k-\tau} - x^\star),\, F_i(\bar{z}_t^k, Y_{i,t}^k) - \bar{F}_i(\bar{z}_t^k)\Big\rangle] \\ + \frac{\alpha}{m}\sum_{i=1}^m \mathbb{E}[\Big\langle \nabla M(\bar{x}^{k-\tau} - x^\star),\, F_i(\bar{x}^{k-\tau}, Y_{i,t}^k) - \bar{F}_i(\bar{x}^{k-\tau})\Big\rangle] \\ + \frac{\alpha}{m}\sum_{i=1}^m \mathbb{E}[\Big\langle \nabla M(\bar{x}^{k-\tau} - x^\star),\, F_i(\bar{z}_t^k, Y_{i,t}^k) - F_i(\bar{x}^{k-\tau}, Y_{i,t}^k)\Big\rangle] \\ + \frac{\alpha}{m}\sum_{i=1}^m \mathbb{E}[\Big\langle \nabla M(\bar{x}^{k-\tau} - x^\star),\, \bar{F}_i(\bar{x}^{k-\tau}) - \bar{F}_i(\bar{z}_t^k)\Big\rangle]\end{aligned} \tag{53}$$

gives the corresponding decomposition of $\mathcal{T}_{k,t}^{\mathrm{mk}}$ into four parts.

$$
\begin{aligned}
&\Big|\Big\langle \nabla M(\bar{z}_t^k - x^\star) - \nabla M(\bar{x}^{k-\tau} - x^\star),\, F_i(\bar{z}_t^k, Y_{i,t}^k) - \bar{F}_i(\bar{z}_t^k)\Big\rangle\Big| \\
&\le \|\nabla M(\bar{z}_t^k - x^\star) - \nabla M(\bar{x}^{k-\tau} - x^\star)\|_s^\star \|F_i(\bar{z}_t^k, Y_{i,t}^k) - \bar{F}_i(\bar{z}_t^k)\|_s \\
&\le L_{ms}\|\bar{z}_t^k - \bar{x}^{k-\tau}\|_s \cdot \frac{1}{\ell_{cs}}\|F_i(\bar{z}_t^k, Y_{i,t}^k) - \bar{F}_i(\bar{z}_t^k)\|_c \\
&\le \frac{L_{ms}}{\ell_{cs}^2}\,\|\bar{z}_t^k - \bar{x}^{k-\tau}\|_c \cdot \|F_i(\bar{z}_t^k, Y_{i,t}^k) - \bar{F}_i(\bar{z}_t^k)\|_c \qquad (54) \\
&\le \frac{2L_{ms}}{\ell_{cs}^2}\,\|\bar{z}_t^k - \bar{x}^{k-\tau}\|_c\,(LR_{k,t} + B), \qquad (55)
\end{aligned}
$$

where the last inequality is due to

$$
\begin{aligned}
&\|F_i(\bar{z}_t^k, Y_{i,t}^k) - \bar{F}_i(\bar{z}_t^k)\|_c \le \big\|F_i(\bar{z}_t^k, Y_{i,t}^k)\big\|_c + \big\|\bar{F}_i(\bar{z}_t^k)\big\|_c \\
&\le \|F_i(\bar{z}_t^k, Y_{i,t}^k) - F_i(x^\star, Y_{i,t}^k)\|_c + \|F_i(x^\star, Y_{i,t}^k)\|_c + \big\|\bar{F}_i(\bar{z}_t^k) - \bar{F}_i(x^\star)\big\|_c + \big\|\bar{F}_i(x^\star)\big\|_c \\
&\le 2(LR_{k,t} + B).
\end{aligned}
$$

Averaging over i yields

$$
\begin{aligned}
&\frac{\alpha}{m}\sum_{i=1}^m \mathbb{E}\Big[\Big\langle \nabla M(\bar{z}_t^k - x^\star) - \nabla M(\bar{x}^{k-\tau} - x^\star),\, F_i(\bar{z}_t^k, Y_{i,t}^k) - \bar{F}_i(\bar{z}_t^k)\Big\rangle\Big] \\
&\le \frac{2\alpha L_{ms}}{\ell_{cs}^2}\,\mathbb{E}[\|\bar{z}_t^k - \bar{x}^{k-\tau}\|_c\,(LR_{k,t} + B)].
\end{aligned} \qquad (56)
$$

$$
\begin{aligned}
&\Big|\mathbb{E}[\Big\langle \nabla M(\bar{x}^{k-\tau} - x^\star),\, F_i(\bar{x}^{k-\tau}, Y_{i,t}^k) - \bar{F}_i(\bar{x}^{k-\tau})\Big\rangle \Big|\, \mathcal{F}_{k-\tau,0}]\Big| \\
&\le \|\nabla M(\bar{x}^{k-\tau} - x^\star)\|_M^\star \Big\|\mathbb{E}[F_i(\bar{x}^{k-\tau}, Y_{i,t}^k) - \bar{F}_i(\bar{x}^{k-\tau})\Big|\, \mathcal{F}_{k-\tau,0}]\Big\|_M \\
&\le \frac{1}{\ell_{cm}^2}\,\|\bar{x}^{k-\tau} - x^\star\|_c \Big\|\mathbb{E}[F_i(\bar{x}^{k-\tau}, Y_{i,t}^k)\Big|\, \mathcal{F}_{k-\tau,0}] - \bar{F}_i(\bar{x}^{k-\tau})\Big\|_c. \qquad (57)
\end{aligned}
$$

By the TV bound (52) and the standard inequality $\|\sum_y (p(y) - \mu(y))g(y)\|_c \le 2\, d_{\mathrm{TV}}(p,\mu)\, \sup_y \|g(y)\|_c$, we have

$$
\Big\|\mathbb{E}[F_i(\bar{x}^{k-\tau}, Y_{i,t}^k)\Big|\, \mathcal{F}_{k-\tau,0}] - \bar{F}_i(\bar{x}^{k-\tau})\Big\|_c \le 2\epsilon \sup_{y\in\mathcal{Y}} \|F_i(\bar{x}^{k-\tau}, y)\|_c \le 2\epsilon\big(L\|\bar{x}^{k-\tau} - x^\star\|_c + B\big). \qquad (58)
$$

Plugging (58) into (57), taking expectation, and averaging over i yields

$$
\begin{aligned}
&\frac{\alpha}{m}\sum_{i=1}^m \mathbb{E}\Big[\Big\langle \nabla M(\bar{x}^{k-\tau} - x^\star),\, F_i(\bar{x}^{k-\tau}, Y_{i,t}^k) - \bar{F}_i(\bar{x}^{k-\tau})\Big\rangle\Big] \\
&\le \frac{2\alpha\epsilon}{\ell_{cm}^2}\,\mathbb{E}[\|\bar{x}^{k-\tau} - x^\star\|_c\big(L\|\bar{x}^{k-\tau} - x^\star\|_c + B\big)].
\end{aligned} \qquad (59)
$$

$$
\begin{aligned}
&\Big|\Big\langle \nabla M(\bar{x}^{k-\tau} - x^\star),\, F_i(\bar{z}_t^k, Y_{i,t}^k) - F_i(\bar{x}^{k-\tau}, Y_{i,t}^k)\Big\rangle\Big| \le \frac{L}{\ell_{cm}^2}\,\|\bar{x}^{k-\tau} - x^\star\|_c\,\|\bar{z}_t^k - \bar{x}^{k-\tau}\|_c, \\
&\Big|\Big\langle \nabla M(\bar{x}^{k-\tau} - x^\star),\, \bar{F}_i(\bar{x}^{k-\tau}) - \bar{F}_i(\bar{z}_t^k)\Big\rangle\Big| \le \frac{L}{\ell_{cm}^2}\,\|\bar{x}^{k-\tau} - x^\star\|_c\,\|\bar{z}_t^k - \bar{x}^{k-\tau}\|_c. \qquad (60)
\end{aligned}
$$

Averaging over i and multiplying by α yields

$$\frac{\alpha}{m}\sum_{i=1}^{m}\mathbb{E}\Big[\Big\langle\nabla M(\bar{x}^{k-\tau}-x^\star),\,F_i(\bar{z}_t^k,Y_{i,t}^k)-F_i(\bar{x}^{k-\tau},Y_{i,t}^k)\Big\rangle+\Big\langle\nabla M(\bar{x}^{k-\tau}-x^\star),\,\bar{F}_i(\bar{x}^{k-\tau})-\bar{F}_i(\bar{z}_t^k)\Big\rangle\Big]$$
$$\le\frac{2\alpha L}{\ell_{cm}^2}\,\mathbb{E}[\|\bar{x}^{k-\tau}-x^\star\|_c\,\|\bar{z}_t^k-\bar{x}^{k-\tau}\|_c]. \tag{61}$$

Combining (56), (59), and (61) gives the following explicit (non-asymptotic) inequality for each inner step t:

$$\mathcal{T}_{k,t}^{\mathrm{mk}}\le\frac{2\alpha\,L_{ms}}{\ell_{cs}^2}\,\mathbb{E}[\|\bar{z}_t^k-\bar{x}^{k-\tau}\|_c\,(LR_{k,t}+B)]+\frac{2\alpha\epsilon}{\ell_{cm}^2}\,\mathbb{E}[\|\bar{x}^{k-\tau}-x^\star\|_c\big(L\|\bar{x}^{k-\tau}-x^\star\|_c+B\big)]$$
$$+\frac{2\alpha L}{\ell_{cm}^2}\,\mathbb{E}[\|\bar{x}^{k-\tau}-x^\star\|_c\,\|\bar{z}_t^k-\bar{x}^{k-\tau}\|_c]. \tag{62}$$

The remaining task is to upper bound the delay drift $\|\bar{z}_t^k-\bar{x}^{k-\tau}\|_c$ using the deterministic recursion.

Lemma 4 (Bound for delay drift)**.** *Consider $\epsilon>0$ and the round-delay τ defined in* (51)*. For any $k\ge\tau, t\in\{0,1,\cdots,H-1\}$, define*

$$\Gamma_{k,\tau}:=\frac{B}{L}+\max_{k-\tau\le q\le k}\max_{0\le s\le H-1}\Big(R_{q,s}+S_{q,s}\Big). \tag{63}$$

Then,

$$\|\bar{z}_t^k-\bar{x}^{k-\tau}\|_c\le\alpha H(\tau+1)(2L+1)\Gamma_{k,\tau}. \tag{64}$$

Proof. Write $\bar{z}_t^k-\bar{x}^{k-\tau}=(\bar{z}_t^k-\bar{x}^k)+(\bar{x}^k-\bar{x}^{k-\tau})$. Then,

$$\|\bar{z}_t^k-\bar{x}^k\|_c=\left\|\sum_{s=0}^{t-1}(\bar{z}_{s+1}^k-\bar{z}_s^k)\right\|_c\le\sum_{s=0}^{t-1}\|\Delta_{k,s}\|_c\le t\cdot\alpha(2L+1)\Gamma_{k,\tau}\le\alpha H(2L+1)\Gamma_{k,\tau}. \tag{65}$$

We need to bound $\|\Delta_{k,s}\|_c$,

$$\frac{1}{\alpha}\|\Delta_{k,s}\|_c=\left\|\frac{1}{m}\sum_{i=1}^{m}F_i(z_{i,s}^k,Y_{i,s}^k)-\bar{z}_s^k\right\|_c$$
$$=\left\|\frac{1}{m}\sum_{i=1}^{m}\big(F_i(z_{i,s}^k,Y_{i,s}^k)-F_i(x^\star,Y_{i,s}^k)\big)+\frac{1}{m}\sum_{i=1}^{m}\big(F_i(x^\star,Y_{i,s}^k)-x^\star\big)+(x^\star-\bar{z}_s^k)\right\|_c$$
$$\le\underbrace{\left\|\frac{1}{m}\sum_{i=1}^{m}\big(F_i(z_{i,s}^k,Y_{i,s}^k)-F_i(x^\star,Y_{i,s}^k)\big)\right\|_c}_{\text{(I)}}+\underbrace{\left\|\frac{1}{m}\sum_{i=1}^{m}\big(F_i(x^\star,Y_{i,s}^k)-x^\star\big)\right\|_c}_{\text{(II)}}+R_{k,s}.$$

For (I),

$$\text{(I)}\le\frac{1}{m}\sum_{i=1}^{m}\|F_i(z_{i,s}^k,Y_{i,s}^k)-F_i(x^\star,Y_{i,s}^k)\|_c\le\frac{L}{m}\sum_{i=1}^{m}\|z_{i,s}^k-x^\star\|_c$$
$$\le\frac{L}{m}\sum_{i=1}^{m}\Big(\|z_{i,s}^k-\bar{z}_s^k\|_c+\|\bar{z}_s^k-x^\star\|_c\Big)=L(S_{k,s}+R_{k,s}).$$

For (II),

$$\text{(II)}\le\frac{1}{m}\sum_{i=1}^{m}\|F_i(x^\star,Y_{i,s}^k)-x^\star\|_c\le B.$$

Combining the above bounds yields

$$\|\Delta_{k,s}\|_c\le\alpha\Big((L+1)R_{k,s}+LS_{k,s}+B\Big)\le\alpha(2L+1)\Gamma_{k,\tau}. \tag{66}$$

For the within-round drift,

$$\|\bar{z}_t^k - \bar{x}^k\|_c = \left\| \sum_{s=0}^{t-1} (\bar{z}_{s+1}^k - \bar{z}_s^k) \right\|_c \le \sum_{s=0}^{t-1} \|\Delta_{k,s}\|_c \le \alpha H(2L+1)\Gamma_{k,\tau}. \tag{67}$$

Then, we consider bounding $\left\|\bar{x}^k - \bar{x}^{k-\tau}\right\|_c$,

$$\bar{x}^k - \bar{x}^{k-\tau} = \sum_{q=k-\tau}^{k-1} (\bar{x}^{q+1} - \bar{x}^q) = \sum_{q=k-\tau}^{k-1} (\bar{z}_H^q - \bar{z}_0^q), \tag{68}$$

and similarly,

$$\|\bar{z}_H^q - \bar{z}_0^q\|_c \le \sum_{s=0}^{H-1} \|\Delta_{q,s}\|_c \le H \cdot \alpha(2L+1)\Gamma_{k,\tau}, \quad \text{for each } q \in [k-\tau, k-1].$$

Therefore,

$$\|\bar{x}^k - \bar{x}^{k-\tau}\|_c \le \sum_{q=k-\tau}^{k-1} \|\bar{z}_H^q - \bar{z}_0^q\|_c \le \tau \cdot \alpha H(2L+1)\Gamma_{k,\tau}. \tag{69}$$

Combining the two parts gives the deterministic bound

$$\|\bar{z}_t^k - \bar{x}^{k-\tau}\|_c \le \alpha H(\tau+1)(2L+1)\Gamma_{k,\tau}. \tag{70}$$

□

Lemma 5 (Markov bias bound for one inner step)**.** *Let $\epsilon = \epsilon(\alpha)$ and the round-delay τ be defined in* (51)*. For any $k \ge \tau$ and $t \in \{0, 1, \ldots, H-1\}$, with $\Gamma_{k,\tau}$ defined in* (63)*, we have*

$$\mathcal{T}_{k,t}^{\mathrm{mk}} \le \alpha^2 H(\tau+1)\,(2L+1)\,L \left(\frac{2L_{ms}}{\ell_{cs}^2} + \frac{2}{\ell_{cm}^2} \right) \mathbb{E}[\Gamma_{k,\tau}^2] + \frac{2\alpha\epsilon L}{\ell_{cm}^2}\, \mathbb{E}[\Gamma_{k,\tau}^2]. \tag{71}$$

Proof. By Lemma 4,

$$\|\bar{z}_t^k - \bar{x}^{k-\tau}\|_c \le \alpha H(\tau+1)(2L+1)\Gamma_{k,\tau}.$$

To bound the remaining factors by $\Gamma_{k,\tau}$, we have

$$\|\bar{x}^{k-\tau} - x^\star\|_c = R_{k-\tau,0} \le \Gamma_{k,\tau}, \qquad R_{k,t} \le \Gamma_{k,\tau}. \tag{72}$$

$$LR_{k,t} + B \le L\Gamma_{k,\tau}. \tag{73}$$

Then

$$\frac{2\alpha\, L_{ms}}{\ell_{cs}^2}\, \mathbb{E}[\|\bar{z}_t^k - \bar{x}^{k-\tau}\|_c\, (LR_{k,t} + B)] \le \frac{2\alpha\, L_{ms}}{\ell_{cs}^2}\, \alpha H(\tau+1)(2L+1)\, L\, \mathbb{E}[\Gamma_{k,\tau}^2]. \tag{74}$$

Using (72)–(73), the second term is bounded by

$$\frac{2\alpha\epsilon}{\ell_{cm}^2}\, \mathbb{E}[\|\bar{x}^{k-\tau} - x^\star\|_c \big(L\|\bar{x}^{k-\tau} - x^\star\|_c + B\big)] \le \frac{2\alpha\epsilon L}{\ell_{cm}^2}\, \mathbb{E}[\Gamma_{k,\tau}^2]. \tag{75}$$

Finally, using (70) and (72), the third term is bounded by

$$\frac{2\alpha L}{\ell_{cm}^2}\, \mathbb{E}[\|\bar{x}^{k-\tau} - x^\star\|_c\, \|\bar{z}_t^k - \bar{x}^{k-\tau}\|_c] \le \frac{2\alpha L}{\ell_{cm}^2}\, \alpha H(\tau+1)(2L+1)\, \mathbb{E}[\Gamma_{k,\tau}^2]. \tag{76}$$

Summing these three bounds yields (71). □

A.4 Analysis of outer loop

We now combine the inner-step descent estimate, the Markovian bias estimate, and the consensus recursion. Throughout this subsection, let $k \geq \tau$, where $\epsilon(\alpha)$ and τ are defined in (51). We use $\Gamma_{k,\tau}$ as defined in (63).

First, we record the inner-loop bounds needed below. Assume

$$\alpha(L+1)H \leq \frac{1}{4}, \qquad \alpha LH \leq \frac{1}{4}. \tag{77}$$

Then Lemma 1 gives

$$\max_{0\leq t\leq H-1} S_{k,t} \leq 2S_{k,0} + 4\alpha H\left(L \max_{0\leq t\leq H-1} R_{k,t} + B\right). \tag{78}$$

Moreover, using

$$\|\bar{z}_{t+1}^k - \bar{z}_t^k\|_c \leq \alpha\left((L+1)R_{k,t} + LS_{k,t} + B\right),$$

we obtain

$$\max_{0\leq t\leq H-1} R_{k,t} \leq R_{k,0} + \alpha H\left((L+1)\max_{0\leq t\leq H-1} R_{k,t} + L\max_{0\leq t\leq H-1} S_{k,t} + B\right).$$

Using $\alpha(L+1)H \leq 1/4$ gives

$$\max_{0\leq t\leq H-1} R_{k,t} \leq \frac{4}{3}\left(R_{k,0} + \alpha H\left(L\max_{0\leq t\leq H-1} S_{k,t} + B\right)\right). \tag{79}$$

Substituting (79) into (78), and then using $\alpha LH \leq 1/4$, yields

$$\max_{0\leq t\leq H-1} S_{k,t} \leq 3S_{k,0} + 8\alpha H\left(LR_{k,0} + B\right). \tag{80}$$

Consequently,

$$\begin{aligned}\sum_{t=0}^{H-1}\mathbb{E}[S_{k,t}^2] &\leq H\mathbb{E}\left[\left(\max_{0\leq t\leq H-1} S_{k,t}\right)^2\right] \\ &\leq 18H\,\mathbb{E}[S_{k,0}^2] + 512\alpha^2H^3L^2u_{cm}^2\,\mathbb{E}[\Phi_{k,0}] + 256\alpha^2H^3B^2.\end{aligned} \tag{81}$$

Here we used $R_{k,0}^2 \leq 2u_{cm}^2\Phi_{k,0}$.

We next sum the one-step potential inequality in Lemma 3. Assume, in addition to (77), that

$$\frac{3L_{ms}(L+1)^2u_{cm}^2}{\ell_{cs}^2}\,\alpha \leq \frac{\kappa}{2}, \qquad \frac{\alpha\kappa H}{2} \leq 1. \tag{82}$$

Then

$$1 - \alpha\kappa + \frac{3L_{ms}\alpha^2(L+1)^2u_{cm}^2}{\ell_{cs}^2} \leq 1 - \frac{\alpha\kappa}{2},$$

and hence

$$\left(1 - \alpha\kappa + \frac{3L_{ms}\alpha^2(L+1)^2u_{cm}^2}{\ell_{cs}^2}\right)^H \leq 1 - \frac{\alpha\kappa H}{4}.$$

Combining Lemma 3, Lemma 5, and (81) gives

$$\begin{aligned}\mathbb{E}[\Phi_{k+1,0}] \leq{}& \left(1 - \frac{\alpha\kappa H}{4}\right)\mathbb{E}[\Phi_{k,0}] \\ &+ \left(\frac{\alpha L^2u_{cm}^2}{2\kappa\ell_{cm}^4} + \frac{3L_{ms}\alpha^2L^2}{2\ell_{cs}^2}\right)\left(18H\,\mathbb{E}[S_{k,0}^2] + 512\alpha^2H^3L^2u_{cm}^2\,\mathbb{E}[\Phi_{k,0}] + 256\alpha^2H^3B^2\right) \\ &+ \left(\alpha^2H^2(\tau+1)(2L+1)L\left(\frac{2L_{ms}}{\ell_{cs}^2} + \frac{2}{\ell_{cm}^2}\right) + \frac{2\alpha\epsilon(\alpha)HL}{\ell_{cm}^2}\right)\mathbb{E}[\Gamma_{k,\tau}^2] + \frac{3L_{ms}\alpha^2H}{2\ell_{cs}^2}B^2.\end{aligned} \tag{83}$$

If also

$$256\frac{\alpha^2 H^2 L^4 u_{cm}^4}{\kappa \ell_{cm}^4} + 768\frac{L_{ms}\alpha^3 H^2 L^4 u_{cm}^2}{\ell_{cs}^2} \le \frac{\kappa}{8}, \tag{84}$$

then the $\mathbb{E}[\Phi_{k,0}]$ contribution generated by the inner-loop disagreement is absorbed into the descent term, and (83) becomes

$$\begin{aligned}\mathbb{E}[\Phi_{k+1,0}] \le &\left(1 - \frac{\alpha\kappa H}{8}\right)\mathbb{E}[\Phi_{k,0}] \\ &+ \alpha H L^2\left(\frac{9u_{cm}^2}{\kappa\ell_{cm}^4} + \frac{27\alpha L_{ms}}{\ell_{cs}^2}\right)\mathbb{E}[S_{k,0}^2] \\ &+ \left(\alpha^2 H^2(\tau+1)(2L+1)L\left(\frac{2L_{ms}}{\ell_{cs}^2} + \frac{2}{\ell_{cm}^2}\right) + \frac{2\alpha\epsilon(\alpha)HL}{\ell_{cm}^2}\right)\mathbb{E}[\Gamma_{k,\tau}^2] \\ &+ \left(\frac{3L_{ms}\alpha^2 H}{2\ell_{cs}^2} + \frac{128\alpha^3 H^3 L^2 u_{cm}^2}{\kappa\ell_{cm}^4} + \frac{384 L_{ms}\alpha^4 H^3 L^2}{\ell_{cs}^2}\right)B^2. \end{aligned}\tag{85}$$

The squared consensus error after one communication also follows from Lemma 2 and (80). Specifically,

$$\begin{aligned}\mathbb{E}[S_{k+1,0}^2] \le &\, 18m\rho^2\left(\frac{u_{2c}}{\ell_{2c}}\right)^2\mathbb{E}[S_{k,0}^2] + 512m\rho^2\left(\frac{u_{2c}}{\ell_{2c}}\right)^2\alpha^2H^2L^2u_{cm}^2\,\mathbb{E}[\Phi_{k,0}] \\ &+ 256m\rho^2\left(\frac{u_{2c}}{\ell_{2c}}\right)^2\alpha^2H^2B^2. \end{aligned}\tag{86}$$

It remains to control the delayed term $\Gamma_{k,\tau}^2$. Assume

$$\tau\alpha LH \le \frac{1}{4}, \qquad \left(\left(1 + \rho\frac{u_{2c}}{\ell_{2c}}\sqrt{m}\right)(1+3\alpha L)^H\right)^\tau \le 2. \tag{87}$$

Using the recursion

$$R_{q,s+1} + S_{q,s+1} \le (1+3\alpha L)(R_{q,s} + S_{q,s}) + 3\alpha B$$

inside each round, together with the communication bound for $S_{q+1,0}$, gives

$$\begin{aligned}\mathbb{E}[\Gamma_{k,\tau}^2] \le &\, 576u_{cm}^2\,\mathbb{E}[\Phi_{k-\tau,0}] + 288\,\mathbb{E}[S_{k-\tau,0}^2] \\ &+ \left(81e^{3/2}\left(1 + \rho\frac{u_{2c}}{\ell_{2c}}\sqrt{m}\right)^2 + 18\right)\frac{B^2}{L^2}. \end{aligned}\tag{88}$$

Moreover, by Lemma 4,

$$\|\bar{x}^k - \bar{x}^{k-\tau}\|_c \le \tau\alpha H(2L+1)\Gamma_{k,\tau}.$$

Thus

$$\Phi_{k-\tau,0} \le 2\Phi_{k,0} + \frac{\tau^2\alpha^2H^2(2L+1)^2}{\ell_{cm}^2}\Gamma_{k,\tau}^2.$$

Taking expectations and using (88), if

$$576u_{cm}^2\frac{\tau^2\alpha^2H^2(2L+1)^2}{\ell_{cm}^2} \le \frac{1}{2}, \tag{89}$$

then

$$\begin{aligned}\mathbb{E}[\Phi_{k-\tau,0}] \le &\, 4\mathbb{E}[\Phi_{k,0}] + 576\frac{\tau^2\alpha^2H^2(2L+1)^2}{\ell_{cm}^2}\mathbb{E}[S_{k-\tau,0}^2] \\ &+ \frac{2\tau^2\alpha^2H^2(2L+1)^2}{\ell_{cm}^2}\left(81e^{3/2}\left(1+\rho\frac{u_{2c}}{\ell_{2c}}\sqrt{m}\right)^2 + 18\right)\frac{B^2}{L^2}. \end{aligned}\tag{90}$$

The final combination uses the following Lyapunov weight:

$$c_S := 4\alpha H L^2 \left(\frac{9u_{cm}^2}{\kappa \ell_{cm}^4} + \frac{27\alpha L_{ms}}{\ell_{cs}^2} \right). \tag{91}$$

For every $k \geq \tau$, define

$$V^k := \mathbb{E}[\Phi_{k,0}] + c_S \mathbb{E}\left[S_{k,0}^2\right] + c_S \mathbb{E}\left[S_{k-\tau,0}^2\right]. \tag{92}$$

We collect the step-size and network conditions used in the final combination. First, assume

$$\alpha(L+1)H \leq \frac{1}{4}, \qquad \alpha L H \leq \frac{1}{4}, \tag{93}$$

$$\frac{3L_{ms}(L+1)^2 u_{cm}^2}{\ell_{cs}^2}\alpha \leq \frac{\kappa}{2}, \qquad \frac{\alpha\kappa H}{2} \leq 1, \tag{94}$$

$$256\frac{\alpha^2 H^2 L^4 u_{cm}^4}{\kappa \ell_{cm}^4} + 768\frac{L_{ms}\alpha^3 H^2 L^4 u_{cm}^2}{\ell_{cs}^2} \leq \frac{\kappa}{8}, \tag{95}$$

$$\tau\alpha L H \leq \frac{1}{4}, \qquad \left(\left(1 + \rho \frac{u_{2c}}{\ell_{2c}} \sqrt{m} \right) (1+3\alpha L)^H \right)^\tau \leq 2, \tag{96}$$

$$576u_{cm}^2 \frac{\tau^2 \alpha^2 H^2 (2L+1)^2}{\ell_{cm}^2} \leq \frac{1}{2}, \tag{97}$$

$$18m\rho^2 \left(\frac{u_{2c}}{\ell_{2c}} \right)^2 \leq \frac{1}{2}. \tag{98}$$

Assume also the final absorption conditions

$$\begin{aligned} &10240m\rho^2 \left(\frac{u_{2c}}{\ell_{2c}} \right)^2 \alpha^3 H^3 L^4 u_{cm}^2 \left(\frac{9u_{cm}^2}{\kappa \ell_{cm}^4} + \frac{27\alpha L_{ms}}{\ell_{cs}^2} \right) \\ &\quad + 4608\alpha^2 H^2 (\tau+1)(2L+1) L u_{cm}^2 \left(\frac{L_{ms}}{\ell_{cs}^2} + \frac{1}{\ell_{cm}^2} \right) + \frac{4608\alpha^2 H L^2 L_{ms} u_{cm}^2 \ell_{cs}^2}{\ell_{cm}^4} \leq \frac{3\alpha\kappa H}{32}, \end{aligned} \tag{99}$$

and

$$\begin{aligned} &288 \left(\alpha^2 H^2 (\tau+1)(2L+1) L \left(\frac{2L_{ms}}{\ell_{cs}^2} + \frac{2}{\ell_{cm}^2} \right) + \frac{2\alpha^2 H L^2 L_{ms} \ell_{cs}^2}{\ell_{cm}^4} \right) \\ &\quad + \frac{576\tau^2 \alpha^2 H^2 (2L+1)^2}{\ell_{cm}^2} \left(512 c_S m \rho^2 \left(\frac{u_{2c}}{\ell_{2c}} \right)^2 \alpha^2 H^2 L^2 u_{cm}^2 \right. \\ &\quad \left. + 576u_{cm}^2 \left(\alpha^2 H^2 (\tau+1)(2L+1) L \left(\frac{2L_{ms}}{\ell_{cs}^2} + \frac{2}{\ell_{cm}^2} \right) + \frac{2\alpha^2 H L^2 L_{ms} \ell_{cs}^2}{\ell_{cm}^4} \right) \right) \leq \frac{c_S}{4}. \end{aligned} \tag{100}$$

Denote

$$\Xi := \frac{48L_{ms}}{\kappa\ell_{cs}^2} + \alpha H^2 L^2 \left(\frac{4096u_{cm}^2}{\kappa^2\ell_{cm}^4} + \frac{12288\alpha L_{ms}}{\kappa\ell_{cs}^2} + \frac{65536m\rho^2\left(\frac{u_{2c}}{\ell_{2c}}\right)^2}{\kappa}\left(\frac{9u_{cm}^2}{\kappa\ell_{cm}^4} + \frac{27\alpha L_{ms}}{\ell_{cs}^2}\right)\right) + \left(81e^{3/2}\left(1+\rho\frac{u_{2c}}{\ell_{2c}}\sqrt{m}\right)^2 + 18\right)\left(\frac{64H\left(\left\lceil \frac{1}{H(1-r_p)}\log\frac{c_p\ell_{cm}^2}{LL_{ms}\ell_{cs}^2\alpha}\right\rceil + 1\right)(2L+1)}{\kappa L}\left(\frac{L_{ms}}{\ell_{cs}^2} + \frac{1}{\ell_{cm}^2}\right) + \frac{64L_{ms}\ell_{cs}^2}{\kappa\ell_{cm}^4} + \alpha^2H^2\left\lceil \frac{1}{H(1-r_p)}\log\frac{c_p\ell_{cm}^2}{LL_{ms}\ell_{cs}^2\alpha}\right\rceil^2(2L+1)^2\left(\frac{131072m\rho^2\left(\frac{u_{2c}}{\ell_{2c}}\right)^2\alpha H^2L^2u_{cm}^2}{\kappa\ell_{cm}^2}\cdot\left(\frac{9u_{cm}^2}{\kappa\ell_{cm}^4} + \frac{27\alpha L_{ms}}{\ell_{cs}^2}\right) + \frac{73728H\left(\left\lceil \frac{1}{H(1-r_p)}\log\frac{c_p\ell_{cm}^2}{LL_{ms}\ell_{cs}^2\alpha}\right\rceil + 1\right)(2L+1)u_{cm}^2}{\kappa L\ell_{cm}^2}\left(\frac{L_{ms}}{\ell_{cs}^2} + \frac{1}{\ell_{cm}^2}\right) + \frac{73728L_{ms}u_{cm}^2\ell_{cs}^2}{\kappa\ell_{cm}^6}\right)\right). \tag{101}$$

Proposition 1 (Round-level Lyapunov recursion for DLSA)**.** *Assume the hypotheses of Lemmas 3–5. If the stepsize and network parameter satisfy* (93)–(100)*, then, for every $k \geq \tau$,*

$$V^{k+1} \leq \left(1 - \frac{\alpha\kappa H}{32}\right)V^k + \frac{\alpha^2\kappa H}{32}B^2\Xi. \tag{102}$$

Proof. Add (85), c_S times (86), and c_S times the shifted version of (86). Then use (88) to replace $\mathbb{E}\left[\Gamma_{k,\tau}^2\right]$, and use (90) to replace $\mathbb{E}[\Phi_{k-\tau,0}]$. The coefficient of $\mathbb{E}[\Phi_{k,0}]$ is at most $1-\alpha\kappa H/32$ by (99). By the definition of c_S, (98), and (94), the coefficient of $\mathbb{E}\left[S_{k,0}^2\right]$ is at most $(1-\alpha\kappa H/32)c_S$. The coefficient of $\mathbb{E}\left[S_{k-\tau,0}^2\right]$ is at most $(1-\alpha\kappa H/32)c_S$ by (98) and (100). In the remaining B^2 terms, substitute $\tau = \left\lceil \frac{1}{H(1-r_p)}\log\frac{c_p\ell_{cm}^2}{LL_{ms}\ell_{cs}^2\alpha}\right\rceil$ and collect the common leading factor $\alpha^2\kappa H/32$. These remaining terms are bounded by $(\alpha^2\kappa H/32)B^2\Xi$ according to (101). This proves (102). □

Corollary 3 (Finite-time convergence and steady-state error)**.** *Under the conditions of Proposition 1, with $R_k := \|\bar{x}^k - x^\star\|_c$ and $S_k := S_{k,0}$, for every $k \geq \tau+1$,*

$$\mathbb{E}\left[R_k^2\right] = \widetilde{\mathcal{O}}\left(\left(1-\frac{\alpha\kappa H}{32}\right)^{k-\tau}\left(\mathbb{E}\left[R_\tau^2\right] + \alpha HL^2\mathbb{E}\left[S_\tau^2\right] + \alpha HL^2\mathbb{E}\left[S_0^2\right]\right)\right) + \widetilde{\mathcal{O}}\left(\alpha B^2\left(1 + \frac{H(2L+1)}{L} + \alpha H^2L^2 + \alpha^2(1+H)^2(2L+1)^2\left(1 + \frac{H(2L+1)}{L} + \alpha H^2L^2\right)\right)\right), \tag{103}$$

$$\mathbb{E}\left[S_k^2\right] = \widetilde{\mathcal{O}}\left(\left(18m\rho^2\left(\frac{u_{2c}}{\ell_{2c}}\right)^2\right)^{k-\tau}\mathbb{E}\left[S_\tau^2\right]\right) + \widetilde{\mathcal{O}}\left(\rho^2\alpha^2H^2L^2\left(1-\frac{\alpha\kappa H}{32}\right)^{k-\tau}\left(\mathbb{E}\left[R_\tau^2\right] + \alpha HL^2\mathbb{E}\left[S_\tau^2\right] + \alpha HL^2\mathbb{E}\left[S_0^2\right]\right)\right) + \widetilde{\mathcal{O}}\left(\rho^2\alpha^2H^2B^2\left(1+\alpha L^2\left(1+\frac{H(2L+1)}{L}+\alpha H^2L^2+\alpha^2(1+H)^2(2L+1)^2\left(1+\frac{H(2L+1)}{L}+\alpha H^2L^2\right)\right)\right)\right). \tag{104}$$

Consequently,

$$\limsup_{k\to\infty} \mathbb{E}\big[R_k^2\big] \tag{105}$$

$$= \widetilde{\mathcal{O}}\Bigg(\alpha B^2\left(1+\frac{H(2L+1)}{L}+\alpha H^2L^2+\alpha^2(1+H)^2(2L+1)^2\left(1+\frac{H(2L+1)}{L}+\alpha H^2L^2\right)\right)\Bigg), \tag{106}$$

$$\limsup_{k\to\infty} \mathbb{E}\big[S_k^2\big] \tag{107}$$

$$= \widetilde{\mathcal{O}}\Bigg(\rho^2\alpha^2H^2B^2\left(1+\alpha L^2\left(1+\frac{H(2L+1)}{L}+\alpha H^2L^2+\alpha^2(1+H)^2(2L+1)^2\left(1+\frac{H(2L+1)}{L}+\alpha H^2L^2\right)\right)\right)\Bigg). \tag{108}$$

Proof. Iterating (102) from τ to k gives

$$V^k \le \left(1-\frac{\alpha\kappa H}{32}\right)^{k-\tau} V^\tau + \alpha B^2\Xi. \tag{109}$$

Since $R_k^2 \le 2u_{cm}^2\Phi_{k,0}$, this implies

$$\mathbb{E}\big[R_k^2\big] \le 2u_{cm}^2\left(1-\frac{\alpha\kappa H}{32}\right)^{k-\tau} V^\tau + 2u_{cm}^2\alpha B^2\Xi. \tag{110}$$

For the consensus error, iterate (86). For every $k \ge \tau+1$,

$$\begin{aligned}\mathbb{E}\big[S_k^2\big] \le{}& \left(18m\rho^2\left(\frac{u_{2c}}{\ell_{2c}}\right)^2\right)^{k-\tau}\mathbb{E}\big[S_\tau^2\big] \\ &+ 512m\rho^2\left(\frac{u_{2c}}{\ell_{2c}}\right)^2\alpha^2H^2L^2u_{cm}^2\sum_{j=\tau}^{k-1}\left(18m\rho^2\left(\frac{u_{2c}}{\ell_{2c}}\right)^2\right)^{k-1-j}\mathbb{E}[\Phi_{j,0}] \\ &+ 256m\rho^2\left(\frac{u_{2c}}{\ell_{2c}}\right)^2\alpha^2H^2B^2\sum_{j=\tau}^{k-1}\left(18m\rho^2\left(\frac{u_{2c}}{\ell_{2c}}\right)^2\right)^{k-1-j}.\end{aligned} \tag{111}$$

Using (109) to upper bound $\mathbb{E}[\Phi_{j,0}]$ and summing the two geometric series gives

$$\begin{aligned}\mathbb{E}\big[S_k^2\big] \le{}& \left(18m\rho^2\left(\frac{u_{2c}}{\ell_{2c}}\right)^2\right)^{k-\tau}\mathbb{E}\big[S_\tau^2\big] \\ &+ \frac{512m\rho^2\left(\frac{u_{2c}}{\ell_{2c}}\right)^2\alpha^2H^2L^2u_{cm}^2}{1-\frac{\alpha\kappa H}{32}-18m\rho^2\left(\frac{u_{2c}}{\ell_{2c}}\right)^2}\left(1-\frac{\alpha\kappa H}{32}\right)^{k-\tau-1}V^\tau \\ &+ \frac{512m\rho^2\left(\frac{u_{2c}}{\ell_{2c}}\right)^2\alpha^3H^2L^2u_{cm}^2B^2\Xi + 256m\rho^2\left(\frac{u_{2c}}{\ell_{2c}}\right)^2\alpha^2H^2B^2}{1-18m\rho^2\left(\frac{u_{2c}}{\ell_{2c}}\right)^2}.\end{aligned} \tag{112}$$

From (101), under the step-size conditions of Proposition 1, and after hiding norm-equivalence constants, κ, L_{ms}, Markov-chain constants, the effective network factor, and logarithmic factors in $\widetilde{\mathcal{O}}$, we have

$$\Xi = \widetilde{\mathcal{O}}\Bigg(1+\frac{H(2L+1)}{L}+\alpha H^2L^2+\alpha^2(1+H)^2(2L+1)^2\left(1+\frac{H(2L+1)}{L}+\alpha H^2L^2\right)\Bigg). \tag{113}$$

Moreover, by the definition of V^τ and c_S,

$$V^\tau = \widetilde{\mathcal{O}}\big(\mathbb{E}\big[R_\tau^2\big]+\alpha HL^2\mathbb{E}\big[S_\tau^2\big]+\alpha HL^2\mathbb{E}\big[S_0^2\big]\big). \tag{114}$$

Substituting (113) and (114) into (110) and (112) gives (103) and (104). Taking $\limsup_{k\to\infty}$ gives (106) and (108). □

Theorem 2 (Simplified sufficient conditions and DLSA convergence)**.** *Let*

$$L_+ := \max\{L, 1\}, \qquad \delta := \frac{u_{2c}}{\ell_{2c}}, \qquad \tau := \left\lceil \frac{1}{H(1-r_p)} \log \frac{c_p \ell_{cm}^2}{L L_{ms} \ell_{cs}^2 \alpha} \right\rceil.$$

There exist positive constants c_α and c_ρ, depending only on the norm-equivalence, envelope, contraction, and Markov-chain constants, such that the conditions

$$\alpha \le c_\alpha \min\left(\frac{1}{HL_+}, \frac{\kappa}{L_+^2}, \frac{\kappa}{HL_+^2}, \frac{\kappa^{1/3}}{H^{2/3} L_+^{4/3}}, \frac{1}{\tau H L_+}, \frac{\kappa}{H(\tau+1)L_+^2} \right) \tag{115}$$

and

$$\rho \le \frac{c_\rho}{\delta \sqrt{m}\, \tau} \tag{116}$$

imply (93)–(100)*. Consequently, for every $k \ge \tau + 1$,*

$$\begin{aligned}
\mathbb{E}\big[R_k^2\big] &= \widetilde{\mathcal{O}}\Bigg(\left(1 - \frac{\alpha\kappa H}{32}\right)^{k-\tau} \left(\mathbb{E}\big[R_\tau^2\big] + \alpha H L_+^2 \mathbb{E}\big[S_\tau^2\big] + \alpha H L_+^2 \mathbb{E}\big[S_0^2\big] \right) \Bigg) \\
&\quad + \widetilde{\mathcal{O}}\big(\alpha B^2 \left(1 + H + \alpha H^2 L_+^2 + \alpha^2 (1+H)^2 L_+^2 \left(1 + H + \alpha H^2 L_+^2\right)\right)\big), \qquad (117) \\
\mathbb{E}\big[S_k^2\big] &= \widetilde{\mathcal{O}}\Big(\left(18 m \rho^2 \delta^2\right)^{k-\tau} \mathbb{E}\big[S_\tau^2\big] \Big) \\
&\quad + \widetilde{\mathcal{O}}\Bigg(\rho^2 \alpha^2 H^2 L_+^2 \left(1 - \frac{\alpha\kappa H}{32}\right)^{k-\tau} \left(\mathbb{E}\big[R_\tau^2\big] + \alpha H L_+^2 \mathbb{E}\big[S_\tau^2\big] + \alpha H L_+^2 \mathbb{E}\big[S_0^2\big] \right) \Bigg) \\
&\quad + \widetilde{\mathcal{O}}\big(\rho^2 \alpha^2 H^2 B^2 \left(1 + \alpha L_+^2 \left(1 + H + \alpha H^2 L_+^2 + \alpha^2 (1+H)^2 L_+^2 \left(1 + H + \alpha H^2 L_+^2\right)\right)\right)\big). \qquad (118)
\end{aligned}$$

In particular,

$$\begin{aligned}
&\limsup_{k\to\infty} \mathbb{E}\big[R_k^2\big] = \widetilde{\mathcal{O}}\big(\alpha B^2 \left(1 + H + \alpha H^2 L_+^2 + \alpha^2 (1+H)^2 L_+^2 \left(1 + H + \alpha H^2 L_+^2\right)\right)\big), \qquad (119) \\
&\limsup_{k\to\infty} \mathbb{E}\big[S_k^2\big] \qquad (120) \\
&= \widetilde{\mathcal{O}}\big(\rho^2 \alpha^2 H^2 B^2 \left(1 + \alpha L_+^2 \left(1 + H + \alpha H^2 L_+^2 + \alpha^2 (1+H)^2 L_+^2 \left(1 + H + \alpha H^2 L_+^2\right)\right)\right)\big). \qquad (121)
\end{aligned}$$

Proof. We first verify that (115) and (116) imply the conditions of Proposition 1. The first term in (115) gives

$$\alpha(L+1)H = O(c_\alpha), \qquad \alpha L H = O(c_\alpha),$$

which implies (93) when c_α is small enough. The second term gives

$$\alpha(L+1)^2 = O(c_\alpha \kappa),$$

and hence implies the first part of (94). The bound $\alpha H L_+ \le O(c_\alpha)$ also gives $\alpha\kappa H \le O(c_\alpha)$ after decreasing c_α, because κ is fixed in the problem constants; hence the second part of (94) holds.

For the round-absorption condition (95), the third and fourth terms in (115) imply

$$\alpha^2 H^2 L^4 = O(c_\alpha^2 \kappa^2), \qquad \alpha^3 H^2 L^4 = O(c_\alpha^3 \kappa),$$

so the two terms on the left-hand side of (95) are at most constant multiples of $c_\alpha^2 \kappa$ and $c_\alpha^3 \kappa$. Thus (95) follows for sufficiently small c_α.

The fifth term in (115) gives

$$\tau \alpha L H \le \tau \alpha H L_+ \le O(c_\alpha), \qquad \tau \alpha H (2L+1) \le O(c_\alpha).$$

Together with (116), this gives

$$\tau \log\left(1 + \rho\delta\sqrt{m}\right) + \tau H \log(1 + 3\alpha L) \le O(c_\rho) + O(c_\alpha).$$

Taking c_α and c_ρ small enough makes the last display at most $\log 2$, which proves the second part of (96); the first part was already shown. The same bound $\tau\alpha H(2L+1) = O(c_\alpha)$ implies (97). Moreover, since $\tau \geq 1$, (116) gives

$$18m\rho^2\delta^2 \leq O(c_\rho^2),$$

which implies (98) for sufficiently small c_ρ.

It remains to check the final absorption conditions. The left-hand side of (99) is bounded, up to fixed problem constants, by

$$\rho^2\alpha^3H^3L_+^4 + \alpha^2H^2(\tau+1)L_+^2 + \alpha^2HL_+^2.$$

Using (116), the first term is no larger than a constant multiple of $c_\rho^2\alpha^3H^3L_+^4/\tau^2$. The third and sixth terms in (115) imply that all three displayed terms are $O(c_\alpha\alpha\kappa H)$. Thus (99) follows after decreasing c_α and c_ρ.

For (100), note that $c_S = \Theta(\alpha HL^2)$ up to fixed problem constants, and the left-hand side is a finite sum of monomials dominated by

$$\alpha^2H^2(\tau+1)L_+^2, \qquad \alpha^2HL_+^2, \qquad \tau^2\alpha^2H^2L_+^2\left(\rho^2c_S\alpha^2H^2L_+^2 + \alpha^2H^2(\tau+1)L_+^2 + \alpha^2HL_+^2\right).$$

The fifth and sixth terms in (115) imply $\tau\alpha HL_+ = O(c_\alpha)$ and $\alpha H(\tau+1)L_+^2 = O(c_\alpha\kappa)$. Together with (116), every monomial in the last display is at most a sufficiently small constant multiple of c_S. Therefore (100) holds.

Thus all conditions of Proposition 1 hold. Applying Corollary 3 gives the finite-time bounds. Finally, in the error terms of Corollary 3, substitute $2L+1 \leq 3L_+$ and use $L_+ \geq 1$. The logarithmic factors coming from the ceiling expression in (101) are absorbed into $\widetilde{\mathcal{O}}$. This yields (117)–(121). □

B Specialization to Decentralized Q-learning

This section verifies that fixed-behavior tabular decentralized Q-learning is a special case of the decentralized stochastic approximation model in Section 3.

B.1 Q-learning as stochastic approximation

Let $d := |\mathcal{S}||\mathcal{A}|$. For each agent $i \in [m]$, the behavior policy π_i and transition kernel P_i induce a Markov chain over state-action pairs with transition matrix

$$K_i\big((s,a),(s',a')\big) = P_i(s' \mid s,a)\pi_i(a' \mid s'). \tag{122}$$

We assume that this chain is irreducible and aperiodic. Let $d_i(s,a)$ denote its stationary distribution. Since the state-action space is finite, there exist constants $c_{p,Q} > 0$ and $r_{p,Q} \in (0,1)$, uniform over agents, such that

$$d_{TV}\left(K_i^\ell((s,a),\cdot), d_i\right) \leq c_{p,Q}r_{p,Q}^\ell, \qquad \forall i \in [m],\ \ell \geq 0. \tag{123}$$

The sample used by Q-learning is

$$Y_i^{(\ell)} = \big(S_i^{(\ell)}, A_i^{(\ell)}, R_i^{(\ell)}, S_i^{(\ell+1)}\big), \qquad R_i^{(\ell)} = r_i(S_i^{(\ell)}, A_i^{(\ell)}).$$

Since $Y_i^{(\ell)}$ is generated from the state-action chain through a fixed transition kernel, the same geometric mixing form holds for the sample chain, possibly after changing $c_{p,Q}$. Thus $r_{p,Q}$ is the Q-learning specialization of the Markov mixing rate r_p in (4).

For a sample $y = (s,a,r,s')$, define

$$F_i(Q,y) := Q + e_{s,a}\Big(r + \gamma_{\text{disc}}\max_{a' \in \mathcal{A}} Q(s',a') - Q(s,a)\Big), \tag{124}$$

where $e_{s,a} \in \mathbb{R}^d$ is the coordinate vector associated with (s,a). Then

$$Q + \alpha(F_i(Q,y) - Q) = Q + \alpha e_{s,a}\Big(r + \gamma_{\text{disc}}\max_{a' \in \mathcal{A}} Q(s',a') - Q(s,a)\Big), \tag{125}$$

which is exactly one sampled tabular Q-learning update at the visited state-action pair. Therefore Algorithm 1 with the operator (124) is decentralized Q-learning with H local sampled Bellman updates between two communication rounds.

For each agent, define the local Bellman optimality operator

$$(T_i Q)(s,a) := r_i(s,a) + \gamma_{\mathrm{disc}} \sum_{s' \in \mathcal{S}} P_i(s' \mid s,a) \max_{a' \in \mathcal{A}} Q(s',a'). \tag{126}$$

Let

$$D_i := \mathrm{diag}\big(d_i(s,a) : (s,a) \in \mathcal{S} \times \mathcal{A}\big), \qquad \bar{d}(s,a) := \frac{1}{m} \sum_{i=1}^{m} d_i(s,a),$$

and

$$\bar{D} := \mathrm{diag}\big(\bar{d}(s,a) : (s,a) \in \mathcal{S} \times \mathcal{A}\big).$$

Assume

$$d_{\min} := \min_{(s,a) \in \mathcal{S} \times \mathcal{A}} \bar{d}(s,a) > 0. \tag{127}$$

Taking expectation of (124) under the stationary sample distribution of agent i gives

$$\bar{F}_i(Q) = \mathbb{E}_{Y \sim \mu_i^{\pi_i}}[F_i(Q,Y)] = Q + D_i(T_i Q - Q). \tag{128}$$

Averaging over agents,

$$\bar{F}_Q(Q) := \frac{1}{m} \sum_{i=1}^{m} \bar{F}_i(Q) = Q + \frac{1}{m} \sum_{i=1}^{m} D_i(T_i Q - Q). \tag{129}$$

Define

$$(\widetilde{T}Q)(s,a) := \frac{\sum_{i=1}^{m} d_i(s,a)(T_i Q)(s,a)}{\sum_{i=1}^{m} d_i(s,a)}. \tag{130}$$

Since $d_{\min} > 0$, the denominator is positive for every (s,a). Then (129) becomes

$$\bar{F}_Q(Q) = Q + \bar{D}(\widetilde{T}Q - Q). \tag{131}$$

The operator $\widetilde{T}$ is a discounted Bellman optimality operator with reward and transition kernel

$$\widetilde{r}(s,a) = \frac{\sum_{i=1}^{m} d_i(s,a) r_i(s,a)}{\sum_{i=1}^{m} d_i(s,a)}, \qquad \widetilde{P}(\cdot \mid s,a) = \frac{\sum_{i=1}^{m} d_i(s,a) P_i(\cdot \mid s,a)}{\sum_{i=1}^{m} d_i(s,a)}.$$

Thus $\widetilde{T}$ is a γ_{disc}-contraction in $\|\cdot\|_\infty$ and has a unique fixed point, denoted by $Q^\star$. Moreover, since $\bar{D}$ is positive diagonal,

$$\bar{F}_Q(Q) = Q \quad \Longleftrightarrow \quad \bar{D}(\widetilde{T}Q - Q) = 0 \quad \Longleftrightarrow \quad \widetilde{T}Q = Q. \tag{132}$$

Hence the fixed point of the averaged stochastic approximation operator $\bar{F}_Q$ is exactly $Q^\star$.

B.2 Verification of Assumption 1

We verify Assumption 1 with

$$\|\cdot\|_c = \|\cdot\|_\infty.$$

For any $y = (s,a,r,s')$ and $Q, Q' \in \mathbb{R}^d$, the sampled operator satisfies

$$\|F_i(Q,y) - F_i(Q',y)\|_\infty \le \|Q - Q'\|_\infty. \tag{133}$$

Also,

$$\begin{aligned} &F_i(Q,y) - Q - F_i(Q',y) + Q' \\ &= e_{s,a} \left(\gamma_{\mathrm{disc}} \left[\max_{a'} Q(s',a') - \max_{a'} Q'(s',a') \right] - (Q(s,a) - Q'(s,a)) \right), \end{aligned} \tag{134}$$

and therefore

$$\|F_i(Q,y) - Q - F_i(Q',y) + Q'\|_\infty \le (1+\gamma_{\mathrm{disc}})\|Q-Q'\|_\infty. \tag{135}$$

Thus Assumption 1(i) holds with

$$L_Q := 1+\gamma_{\mathrm{disc}}. \tag{136}$$

Next, using (131),

$$\bar{F}_Q(Q) - \bar{F}_Q(Q') = (I-\bar{D})(Q-Q') + \bar{D}(\widetilde{T}Q - \widetilde{T}Q').$$

Since $\widetilde{T}$ is a γ_{disc}-contraction in $\|\cdot\|_\infty$, for every (s,a),

$$\begin{aligned}
|(\bar{F}_Q(Q) - \bar{F}_Q(Q'))(s,a)| &\le (1-\bar{d}(s,a))\|Q-Q'\|_\infty + \bar{d}(s,a)\gamma_{\mathrm{disc}}\|Q-Q'\|_\infty \\
&= \big(1-(1-\gamma_{\mathrm{disc}})\bar{d}(s,a)\big)\,\|Q-Q'\|_\infty \\
&\le (1-(1-\gamma_{\mathrm{disc}})d_{\min})\,\|Q-Q'\|_\infty.
\end{aligned} \tag{137}$$

Therefore Assumption 1(ii) holds with

$$\beta_Q := 1-(1-\gamma_{\mathrm{disc}})d_{\min}. \tag{138}$$

Finally, since $Q^\star$ is the fixed point of a discounted Bellman optimality operator with rewards bounded by $R_{\max}$,

$$\|Q^\star\|_\infty \le \frac{R_{\max}}{1-\gamma_{\mathrm{disc}}}. \tag{139}$$

For any sample $y=(s,a,r,s')$,

$$\|F_i(Q^\star,y)\|_\infty \le \max\left\{\|Q^\star\|_\infty,\, R_{\max}+\gamma_{\mathrm{disc}}\|Q^\star\|_\infty\right\} \le \frac{R_{\max}}{1-\gamma_{\mathrm{disc}}}.$$

Thus Assumption 1(iii) holds with

$$B_Q := \frac{2R_{\max}}{1-\gamma_{\mathrm{disc}}}. \tag{140}$$

B.3 Moreau-envelope constants for the ℓ_∞ contraction

The Moreau-envelope analysis requires the constants relating $\|\cdot\|_\infty$ and the Moreau-envelope norm $\|\cdot\|_M$. Following the generalized Moreau-envelope construction, choose

$$p_Q := \max\{2, 2\log d\}, \qquad \|\cdot\|_s := \|\cdot\|_{p_Q}. \tag{141}$$

Then

$$d^{-1/p_Q}\|x\|_{p_Q} \le \|x\|_\infty \le \|x\|_{p_Q}, \qquad \forall x\in\mathbb{R}^d. \tag{142}$$

Thus

$$\ell_{cs,Q} = d^{-1/p_Q}, \qquad u_{cs,Q} = 1. \tag{143}$$

Take

$$\theta_Q := \left(\frac{1+\beta_Q}{2\beta_Q}\right)^2 - 1. \tag{144}$$

By the generalized Moreau-envelope norm equivalence,

$$\ell_{cm,Q} = \left(1+\theta_Q \ell_{cs,Q}^2\right)^{1/2} = \left(1+\theta_Q d^{-2/p_Q}\right)^{1/2}, \tag{145}$$

and

$$u_{cm,Q} = \left(1+\theta_Q u_{cs,Q}^2\right)^{1/2} = (1+\theta_Q)^{1/2} = \frac{1+\beta_Q}{2\beta_Q}. \tag{146}$$

The smoothness constant of the envelope may be taken as

$$L_{ms,Q} := \frac{p_Q - 1}{\theta_Q}. \tag{147}$$

Therefore the contraction gap in the Moreau-envelope Lyapunov recursion is

$$\kappa_Q := 1 - \beta_Q \frac{u_{cm,Q}}{\ell_{cm,Q}} = 1 - \beta_Q \left(\frac{1+\theta_Q}{1+\theta_Q d^{-2/p_Q}} \right)^{1/2}. \tag{148}$$

Equivalently,

$$\kappa_Q = 1 - \frac{1+\beta_Q}{2} \left(1 + d^{-2/p_Q} \left[\left(\frac{1+\beta_Q}{2\beta_Q} \right)^2 - 1 \right] \right)^{-1/2}. \tag{149}$$

When $d \geq 3$, $p_Q = 2\log d$, and hence $d^{-2/p_Q} = e^{-1}$. In this common case,

$$\kappa_Q = 1 - \frac{1+\beta_Q}{2} \left(1 + \frac{1}{e} \left[\left(\frac{1+\beta_Q}{2\beta_Q} \right)^2 - 1 \right] \right)^{-1/2}. \tag{150}$$

B.4 Applying the DLSA theorem

We now apply Theorem 1. In the Q-learning specialization, the generic quantities R_k and S_k in Theorem 1 are read under the substitutions

$$x_i^k = Q_i^k, \qquad x^\star = Q^\star, \qquad \|\cdot\|_c = \|\cdot\|_\infty.$$

Thus the theorem controls the squared estimation error of the averaged Q-table and the squared consensus error among local Q-tables.

Define the Q-learning Markov delay

$$\tau_Q := \left\lceil \frac{1}{H(1-r_{p,Q})} \log \frac{c_{p,Q}\ell_{cm,Q}^2}{L_Q L_{ms,Q} \ell_{cs,Q}^2 \alpha} \right\rceil. \tag{151}$$

Then

$$\tau_Q = \widetilde{\mathcal{O}} \left(\frac{1}{H(1-r_{p,Q})} \right). \tag{152}$$

Let ρ_{eff} denote the effective network mixing factor in one communication round. For plain DLSA, $\rho_{\text{eff}} = \rho$; for DLSA-FastMix, $\rho_{\text{eff}} = \rho_{\text{fm}}$. If

$$\alpha \leq \frac{c_{\alpha,Q}}{H(\tau_Q+1)(1+L_Q)^2}, \qquad \rho_{\text{eff}} \leq \frac{c_{\rho,Q}}{\sqrt{m}(\tau_Q+1)} \tag{153}$$

for sufficiently small problem-dependent constants $c_{\alpha,Q}, c_{\rho,Q} > 0$, then Theorem 1 gives, for all $k \geq \tau_Q$,

$$\begin{aligned} \mathbb{E}\big[R_k^2\big] &\leq C_{R,Q} \left(1 - \frac{\alpha\kappa_Q H}{32} \right)^{k-\tau_Q} \left(\mathbb{E}\Big[R_{\tau_Q}^2\Big] + \mathbb{E}\Big[S_{\tau_Q}^2\Big] + \mathbb{E}\big[S_0^2\big] \right) \\ &\quad + \widetilde{\mathcal{O}} \left(\alpha B_Q^2 \left(1 + H + \alpha H^2 (1+L_Q)^2 + \alpha^2 H^3 (1+L_Q)^2 \right) \right), \end{aligned} \tag{154}$$

$$\begin{aligned} \mathbb{E}\big[S_k^2\big] &\leq C_{S,Q} \left(1 - \frac{\alpha\kappa_Q H}{32} \right)^{k-\tau_Q} \left(\mathbb{E}\Big[R_{\tau_Q}^2\Big] + \mathbb{E}\Big[S_{\tau_Q}^2\Big] + \mathbb{E}\big[S_0^2\big] \right) \\ &\quad + \widetilde{\mathcal{O}} \left(\rho_{\text{eff}}^2 \alpha^2 H^2 B_Q^2 \left[1 + \alpha (1+L_Q)^2 \left(1 + H + \alpha H^2 (1+L_Q)^2 + \alpha^2 H^3 (1+L_Q)^2 \right) \right] \right). \end{aligned} \tag{155}$$

Here $C_{R,Q}, C_{S,Q} > 0$ are problem-dependent constants, and the hidden constants in $\widetilde{\mathcal{O}}(\cdot)$ depend only on fixed norm-equivalence, envelope, Markov-chain, and contraction constants.

Corollary 1 now applies with the substitutions

$$L = L_Q, \qquad B = B_Q, \qquad \tau = \tau_Q, \qquad c_0 = \frac{\kappa_Q}{32}.$$

For a target accuracy $\varepsilon \in (0,1)$, choose an admissible constant stepsize satisfying

$$\alpha = \min \left\{ \frac{c_{\alpha,Q}}{H(\tau_Q+1)(1+L_Q)^2}, \frac{\varepsilon}{B_Q^2(1+H)}, \frac{\sqrt{\varepsilon}}{B_Q H} \right\}. \tag{156}$$

Then after

$$K = \widetilde{\mathcal{O}}\left(\tau_Q + \frac{1}{\alpha\kappa_Q H}\log\frac{1}{\varepsilon}\right) \tag{157}$$

communication rounds,

$$\mathbb{E}\big[R_K^2\big] \le \widetilde{\mathcal{O}}(\varepsilon), \qquad \mathbb{E}\big[S_K^2\big] \le \widetilde{\mathcal{O}}(\varepsilon). \tag{158}$$

The corresponding local sample complexity per agent is

$$KH = \widetilde{\mathcal{O}}\left(\tau_Q H + \frac{1}{\alpha\kappa_Q}\log\frac{1}{\varepsilon}\right). \tag{159}$$

Equivalently, substituting (156) gives the communication complexity

$$\widetilde{\mathcal{O}}\left(\tau_Q + \max\left\{\frac{(\tau_Q+1)(1+L_Q)^2}{\kappa_Q}, \frac{B_Q^2(1+H)}{\kappa_Q H\varepsilon}, \frac{B_Q}{\kappa_Q\sqrt{\varepsilon}}\right\}\log\frac{1}{\varepsilon}\right), \tag{160}$$

and the local sample complexity

$$\widetilde{\mathcal{O}}\left(\tau_Q H + \max\left\{\frac{H(\tau_Q+1)(1+L_Q)^2}{\kappa_Q}, \frac{B_Q^2(1+H)}{\kappa_Q\varepsilon}, \frac{B_Q H}{\kappa_Q\sqrt{\varepsilon}}\right\}\log\frac{1}{\varepsilon}\right). \tag{161}$$

In particular, when H, L_Q, B_Q are fixed and $\alpha = \Theta(\varepsilon)$ is admissible, the leading-order communication and sample complexities reduce to

$$\widetilde{\mathcal{O}}\left(\tau_Q + \frac{1}{\kappa_Q H\varepsilon}\log\frac{1}{\varepsilon}\right) \quad \text{and} \quad \widetilde{\mathcal{O}}\left(\tau_Q H + \frac{1}{\kappa_Q\varepsilon}\log\frac{1}{\varepsilon}\right), \tag{162}$$

respectively.

C Additional Experiments

We also implement Linear SA and SGD with Markovian noise.

C.1 Linear SA

We first consider a decentralized linear-SA / quadratic-optimization problem with network-average objective

$$f(x) = \frac{1}{m}\sum_{i=1}^{m} f_i(x),$$

where each local loss $f_i(x) = \mathbb{E}[\ell_i(x; Y_{i,t}^k)]$ is accessed only through a stream of temporally correlated Markovian samples $\{Y_{i,t}^k\}$. Each agent forms a stochastic gradient $g_i(x; Y_{i,t}^k)$ and aims to converge to the unique minimizer $x^\star$. The stepsize is constant within each run but tuned separately for each H. With a target error level, we select a personalized stepsize α_H via binary search and use the largest α_H satisfying this criterion. We report

$$\mathrm{err}_k = \frac{1}{m}\sum_{i=1}^{m}\|x_i^k - x^\star\|_2$$

at each communication round. Here each communication round consists of H local stochastic gradient steps $z_{i,t+1}^k = z_{i,t}^k - \alpha_H g_i(z_{i,t}^k; Y_{i,t}^k)$ for $t = 0, \dots, H-1$, followed by one consensus step $x^{k+1} \leftarrow W z_H^k$ using a symmetric doubly-stochastic mixing matrix W on a connected peer-to-peer graph. We measure communication cost by the number of times W is applied,

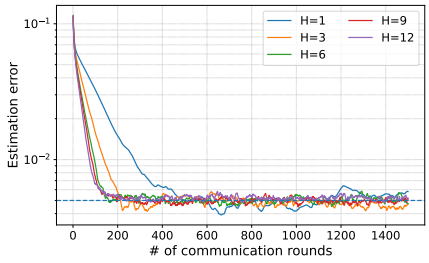


Figure 4: Estimation error versus communication rounds for different numbers of local updates H in the linear-SA experiment. The blue dashed line denotes the target precision, which is set to 5×10^{-3}.

which cleanly separates the effect of the local-update parameter H in error-versus-communication plots. Here we fix the number of agents to $m = 10$ and the dimension to $d = 50$. We consider $H \in \{1, 3, 6, 9, 12\}$ and run each method for $K = 1500$ communication rounds.

This result in Figure 4 highlights the role of local computation: performing more local updates between communication rounds lets each round make larger progress, thereby saving communication while achieving the same target accuracy over the tested range of H.

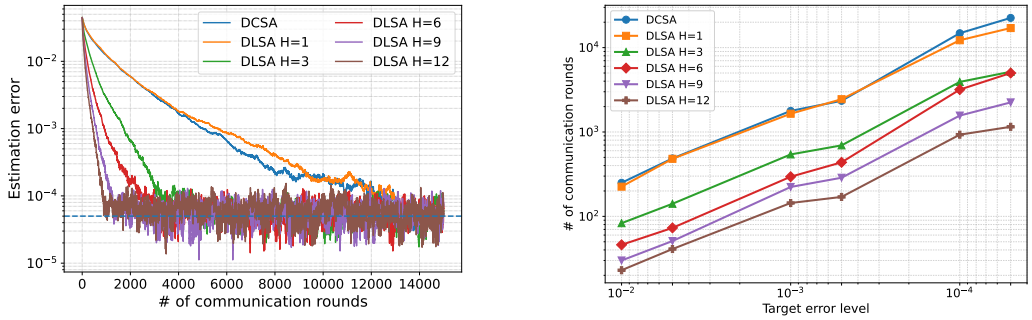
C.2 Decentralized local SGD with Markovian samples

To better evaluate the effectiveness of DLSA, we construct a decentralized system-identification problem with a synthetic Markovian data model. We initialize $m = 50$ agents, each maintaining a $d = 20$-dimensional Markov state $\phi_{i,t} \in \mathbb{R}^d$. For each agent i, at time t we draw $u_{i,t} \sim \mathcal{N}(0, 1)$, set the first coordinate $\phi_{i,t}(1) = u_{i,t}$, and update the remaining coordinates via the subdiagonal recursion

$$\phi_{i,t}(r) = a_{i,r}\, \phi_{i,t-1}(r-1), \qquad r = 2, \dots, d,$$

where $a_{i,r} \sim \mathrm{Unif}[0.8, 0.99]$. A global unknown parameter $x^\star$ is sampled uniformly from the ℓ_2 unit ball, and each agent forms squared-loss stochastic gradients from the resulting Markovian observations. Communication is defined on a line graph using a lazy Metropolis mixing matrix W.

Given this streaming Markov data, we compare DLSA with different H values against the $H = 1$ baseline DCSA [38] to estimate $x^\star$. For each method and each H, we automatically select a stable constant stepsize via a feasibility search so that all methods reach a common target steady-state error. The implementation tracks performance by forming the network average $\bar{x}^k = \frac{1}{m}\sum_{i=1}^m x_i^k$ after each communication round and recording the estimation error $\|\bar{x}^k - x^\star\|_2$.



(a) Estimation error versus communication rounds for different numbers of local updates H for Markovian SGD. The blue dashed line denotes the target precision, which is set to 5×10^{-5}.

(b) Communication Rounds Required to Reach a Target Estimation Error.

Figure 5: Communication–Accuracy Tradeoff under Different Numbers of Local Updates for Markovian SGD

The results in Figure 5(a) show that when $H = 1$, DCSA performs similarly to DLSA, but as H increases, the advantage of doing more local updates becomes apparent. DLSA achieves the same accuracy with fewer communication rounds, thereby significantly reducing the communication cost. Besides, we also report the number of communication rounds needed to reach each target error level. The curves compare the baseline DCSA with DLSA using different numbers of local updates $H \in \{1, 3, 6, 9, 12\}$. For each method and each H, we use the same stepsize-selection procedure and report the smallest round index k such that the target error is achieved.

As shown in Figure 5(b), local SA requires fewer consensus operations than DCSA to reach the same target accuracy, and the gap widens as the target becomes more stringent. While DCSA effectively pays a communication cost at every noisy update, local SA amortizes communication by performing $H > 1$ local SGD steps before mixing. Over the tested range of H, this leads to a better communication–accuracy tradeoff. More broadly, however, the theory does not predict that larger H is always better: overly large local-update budgets or poorly tuned stepsizes can increase disagreement and worsen the eventual error floor.

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